MPhil Thesis

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Contents

1 Introduction

Governments face a key tension in pharmaceutical policy. Public procurement systems and insurance programmes can reduce drug prices and lower state spending, but they may also lower the expected returns to research and development (R&D), weakening incentives for pharmaceutical innovation. This tension arises due to novel drug development requiring large upfront investments over a long horizon, including patent applications and clinical trials, with a high probability of failure (Sutton, 1998). The returns to a successful drug depend on both prices and volumes and accrue over multiple years. Incentives to invest are therefore shaped not only by clinical opportunity but also by the economic and policy settings which influence the expected revenues that can be generated once a drug reaches the market. Garthwaite and Starc frame this as a broader challenge within pharmaceutical pricing reforms, emphasising that policy-makers must balance improved healthcare access through lower drug spending against the incentives required to maintain drug development.

This tension makes the pharmaceutical market within the United States (US) an important setting for studying how public policy shapes innovation incentives. Federal and state governments are major players in the market, with Medicaid and Medicare together accounting for approximately 40% of total spending among the ten top-selling prescription drugs in the US between 2017 and 2019 (Chavehpour et al., 2024). Medicaid is especially important in this respect - it is a large public buyer, but it does not operate as a standard private insurer. Manufacturers that wish to have Medicaid coverage must participate in the Medicaid Drug rebate Program, and states can use Preferred Drug Lists (PDLs) in order to negotiate additional rebates in exchange for preferred formulary placement. Medicaid demand therefore passes through a public procurement system rather than a conventional insurance market, and its institutional features are described in detail in Section 2.

This thesis examines the consequences of a major expansion of a US public insurance programme for pharmaceutical innovation - the 2014 Medicaid expansion under the Affordable Care Act (ACA). The reform produced a well-documented expansion in the size of the Medicaid market, with population coverage increasing by close to twenty million enrollees in the years following the policy change (KFF, 2026). Under standard market-sized frameworks in the pharmaceutical sector this should increase innovation incentives for drug-classes more exposed to Medicaid demand (Finkelstein, 2004; Dubois et al., 2015). If firms expect to sell more units after a successful round of innovation, the expected returns to R&D rise. However, the additional demand entered the market through Medicaid's procurement institutions rather than through standard private channels, so an expansion of Medicaid was not simply a positive demand shock for pharmaceutical firms.

The Medicaid setting complicates the prediction from standard pharmaceutical market-sized frameworks. The ACA did not only expand the number of patients covered by Medicaid, but it also raised the statutory rebate percentages and extended rebate obligations to Medicaid managed care. Therefore, the reform increased the size of the public market while also lowering the share of revenue retained by manufacturers. Additionally, as Medicaid demand grows, preferred placement on a state PDL becomes more valuable. States can use this to negotiate deeper rebates from manufacturers. The same policy may therefore generate two opposing effects: it raises demand, while also increasing pricing pressure and reducing the resulting surplus that firms can capture from that demand (Duggan and Scott Morton, 2006a; Feng et al., 2023; Garthwaite and Starc, 2023).

Public programmes are increasingly central to the pharmaceutical market in the United States. A large share of prescription drug spending is financed by the public sector, and recent reforms have placed greater emphasis on using public purchasing power to reduce drug prices. The 2022 Inflation Reduction Act is a clear example of this shift, where Medicare price negotiation have been introduced for high-spend drugs. These reforms raise the same broad policy tension studied in this thesis: lower prices can benefit patients and taxpayers in the short run, but they may also affect the expected returns to future pharmaceutical innovation.

The Medicaid expansion provides a useful benchmark for this wider policy question by combining a large coverage expansion with an established system of public reimbursement and rebate negotiation. Similar questions arise wherever public buyers combine broad coverage with price control, including the NHS in the United Kingdom and other publicly financed health systems.

The implication is not that public bargaining is necessarily harmful. Rather, the point is that public reimbursement rules can affect pharmaceutical innovation incentives, not only current drug spending (Lakdawalla and Sood, 2009; Lakdawalla, 2018). This thesis studies that relationship in the context of Medicaid, where public demand expanded while the institutions governing public reimbursement became more important.

This thesis therefore raises the question of whether the ACA-induced Medicaid expansion raised pharmaceutical innovation, and whether the response dependent on a firm's exposure to the Medicaid procurement institutions. Put simply, was the expected positive relationship between market size and innovation weakened when the additional demand flows through a state-wide monopsony?

This thesis makes several contributions. First, it constructs a firm-by-therapeutic-class panel linking patent records, FDA approvals, Medicaid prescription data, and PDL placement over 2005-2025, with the empirical analysis using observations from 2010 onwards. This exercise is non-trivial since the underlying sources do not share common firm, drug, or therapeutic-class identifiers. Second, it studies the innovation response to ACA-induced Medicaid expansion using patents and FDA approvals rather than clinical trial activity, while extending the post-expansion period through to 2025. Third, it moves beyond aggregate therapeutic-class exposure by linking innovation outcomes to firm-level exposure to Medicaid's PDL, allowing the analysis to separate Medicaid demand exposure from PDL exposure and to examine whether responses vary across therapeutic areas. Fourth, it allows the innovation response to vary across ATC therapeutic areas, rather than imposing a single average response across all drug classes. Finally, the thesis conducts PDL counterfactual analysis and builds a theoretical model of firm innovation behaviour to interpret the empirical results and show how an expansion in Medicaid demand can be offset by the Medicaid monopsony wedge.

The main empirical finding of this thesis is that therapeutic classes with greater pre-expansion Medicaid exposure experienced a decline in patenting following the 2014 ACA Medicaid expansion, while the evidence for FDA approvals appears to be inconsistent. This suggests that the innovation response to the policy was concentrated in earlier stages of the R&D pipeline rather than at the final point of regulatory approval. Accounting for firm-level exposure to Medicaid's pricing and formulary systems reveals that firms with greater exposure to Medicaid's preferred drug list experience stronger declines in patenting, consistent with the pricing channel reducing the returns to innovation. The negative patenting response is most pronounced in classes where Medicaid exposure is high and products are more substitutable, while other classes show weaker or more positive responses. The theoretical model provides an interpretation for these patterns: while the Medicaid expansion raised expected demand, it also increased a procurement wedge faced by firms through rebates and bargaining. If the procurement effect is large enough, then even while Medicaid demand increases, firms may not expect to earn sufficient profits for innovation incentives to rise. This lends an explanation for why the standard market-size prediction may fail in this setting.

The closest related paper is Garthwaite et al. (2021), which studies clinical trial activity around the Medicaid expansion and finds little evidence of an aggregate innovation response. They propose that the inconclusive results may reflect lower returns associated with Medicaid's pricing rules, but do not test this channel directly. This thesis builds on their work by extending the outcome set to patents and FDA approvals, which bracket clinical trials in the development pipeline, thereby capturing earlier and later stage innovation activity. It also extends the post-expansion window to 2025, and moves from therapeutic-class-level analysis to a firm-class panel. Importantly, the analysis directly examines the pricing channel proposed in their interpretation by linking firm-therapeutic class exposure to Medicaid's formulary system. The empirical results are then used to inform the illustrative model developed later in the thesis, which studies how the Medicaid expansion affects firm innovation incentives when the preferential status on the PDL changes the surplus firms expect to capture.

The remainder of the thesis is structured as follows. Section 2 provides the institutional background needed for the empirical analysis, setting out the ACA Medicaid expansion, the Medicaid rebate system, and the Preferred Drug List mechanism in greater detail. Section 3 reviews the related literature on market size and pharmaceutical innovation, firm incentives under Medicaid's pricing regime, and the pharmaceutical innovation pipeline. Section 4 describes the data sources and the construction of the main variables, including Medicaid demand exposure and PDL exposure at the firm-by-class level. Section 5 sets out the empirical strategy for the three research questions. Section 6 reports the empirical results. Section 7 presents robustness checks. Section 8 presents the theoretical model, including the structure of the game, the mapping from theoretical states to empirical outcomes, the equilibrium analysis, and the comparative statics. Section 9 discusses the policy implications, focusing on the tension between lower net drug prices and the innovation incentives required to justify upfront R&D investment. Section 10 discusses extensions and directions for further research with section 11 concluding the thesis.

2 The Affordable Care Act

Medicaid is a federal-state public health insurance programme for low-income individuals and other eligible groups in the United States. Prior to 2014 eligibility was largely restricted to specific categories, including low-income children, pregnant women, disabled individuals, and some elderly individuals, as opposed to low-income adults in general.

While prescription drug benefits were formally non-mandatory, all state Medicaid programmes provided them. Additionally, outpatient prescriptions are reimbursed through fee-for-service and managed care arrangements.

The ACA substantially expanded this market. From 2014, states were allowed to extend Medicaid eligibility to adults with incomes up to 138% of the federal poverty line. This threshold change generated a large increase in Medicaid enrolment, with adult enrolment increasing by 20.3 million as of January 2025, of whom 16.8 million were newly eligible under the reform KFF (2026). This expansion in coverage corresponding increase in the number of prescriptions reimbursed through Medicaid.

Importantly, there are two key institutional features of Medicaid that are noteworthy. The first is the Medicaid Drug Rebate Program (MDRP). Manufacturers that want their drugs to be reimbursed by Medicaid must enter into a national rebate agreement and pay rebates to states for each unit dispensed to Medicaid. Specifically, for brand-name drugs, the basic rebate applied is based on what is greater between a fixed percentage of the average manufacturer price (AMP), or the difference between AMP and the manufacturer's lowest commercial price, known as the "best price". Consequently, discounts offered elsewhere in the market can affect the rebates owed to Medicaid (Duggan and Scott Morton, 2006a). Medicaid therefore receives a legally protected discount thereby lowering the value of Medicaid prescriptions to manufacturers.

The ACA also changed the rebate environment directly. The statutory rebate percentage under the MDRP was increased from 15.1% to 23.1% of AMP for most brand-name drugs, and from 11% to 13% for generics. Firms were now obligated to extend rebates to drugs dispensed through Medicaid managed care organisations, while previously being exempt.

The second institutional feature is the Preferred Drug List (PDL). States can negotiate supplemental rebates from manufacturers in exchange for preferred placement on their formulary. Preferred drugs are prescribed to patients without prior authorisation, while non-preferred drugs require additional approval before being prescribed. Requesting authorisation creates an administrative cost for physicians, shifting prescribing behaviour towards preferred drugs (Virabhak and Sohn, 2009), indicating that preferred placement can affect a firm's exposure to Medicaid. Preferred placement therefore provides states with a source of negotiating leverage. Since preferred status affects access conditions and expected utilisation within a therapeutic class, manufacturers may offer supplemental rebates above the statutory MDRP minimum in order to obtain or retain preferred placement. Additionally, some states also participate in multi-state purchasing alliances, which aggregate Medicaid demand across states and may further strengthen their bargaining position.

Selection into the PDL takes on the form of a two-stage selection mechanism. In the first stage, the state's Pharmacy and Therapeutics committee assesses whether a drug is clinically appropriate for preferred status. This involves the use of clinical evidence from peer-reviewed clinical literature and oncology reports to understand the safety, efficacy and effectiveness of the drug relative to other drugs with the same role. In the second stage, once drugs pass the first stage and are concluded to be therapeutically comparable, states consider the net cost and supplemental rebates when deciding which drugs receive preferred placement. This setting matters as it means that Medicaid procurement does not only depend on clinical value, but also on firms' willingness to offer price concessions.

For this reason, an expansion of Medicaid demand may also increase the value of preferred placement and intensify competition over rebates.

Importantly, preferred placement on the PDL is not necessarily limited to a single drug for a given drug role. States may allow for multiple preferred drugs within the same role in order to allow for clinically appropriate choices among patients, such as in the case of route of administration, prior treatment response or clinical needs such as allergies. Furthermore, drugs with non-preferred status are not necessarily excluded from Medicaid coverage, they would instead require authorisation from Medicaid once a physician requests to prescribing them.

Thus, the ACA created a demand shock, but it did so inside a public procurement system with rebates and formulary bargaining. For high-Medicaid therapeutic classes, the reform therefore increased the number of prescriptions likely to be paid for by Medicaid, while also increasing the importance of the rebate and PDL mechanisms that determine the net revenue firms receive from those prescriptions.

This thesis does not utilise the staggered timing of state adoption as the main source of variation. Although that variation is useful for studying outcomes such as insurance coverage, healthcare use, or labour supply, it is less suited towards studying pharmaceutical innovation. Pharmaceutical innovation is a national investment decision, and firms cannot develop a drug for sale in one expansion state alone. Instead, they consider expected sales across the United States. Therefore, this thesis treats the start of the ACA Medicaid expansion in 2014 as the relevant policy shock.

3 Literature Review

This thesis sits at the intersection of the literature on pharmaceutical innovation and on how the institutional settings behind public reimbursement shape firm incentives. It also draws on work examining the timing of innovation measures across the drug-development pipeline.

Before turning to these areas, it is beneficial to understand the current studies on the ACA Medicaid expansion. The reform yielded large gains in coverage, concentrated in states that chose to expand access to Medicaid, with the gaps in racial and ethnic coverage narrowing as a result (Courtemanche et al., 2020; Frean et al., 2017; Miller and Wherry, 2019). These states saw increased affordability, indicating reductions in the cost barriers to accessing healthcare (Miller and Wherry, 2017), alongside declines in medical debt and healthcare-related bankruptcy (Brevoort et al., 2017; Hu et al., 2018). These increases in access and financial welfare were accompanied by improvements in self-reported health and reductions in adult mortality (Miller et al., 2021; Miller and Wherry, 2017). However, evidence on labour market effects has been more limited, with no significant changes recorded in employment outcomes or physician hours (Duggan et al., 2019; Dague et al., 2017; Neprash et al., 2021). This thesis studies a less explored dimension of the Medicaid expansion - its effects on pharmaceutical innovation.

3.1 Market Size and Pharmaceutical Innovation

A key prediction in the economics of innovation is that positive market size shocks raise the expected returns to R&D and therefore stimulate innovation Dasgupta and Stiglitz (1980), with the broader literature within industrial organisation predicting that larger markets create incentives for firms to incur endogenous sunk investments, including R&D expenditure (Sutton and Chandler, 1991; Sutton, 1998). In pharmaceutical markets specifically, Sutton (1998) characterises the drug discovery process as a costly lottery with large fixed project costs, where larger expected returns may incentivise firms to enter the R&D race for therapeutic molecules. Civan and Maloney (2006) finds that the number of drugs in the development pipeline is positively associated with disease burden in the United States. Subsequent causal work has strengthened this finding. Acemoglu and Linn (2004) exploit variation in future expected market size based on US demographic trends and find that a one percent increase in potential market size leads to a four percent increase in novel drug approvals. Similarly, Finkelstein (2004) finds that policies aimed at raising vaccine utilisation led to a 250% increase in vaccine-related clinical trials, indicating that innovation responds positively to shocks in expected returns, which is consistent with Dubois et al. (2015), who estimate a positive elasticity of innovation to expected market size in a cross-country setting. Blume-Kohout and Sood (2013) ties this framework to a policy setting, where they find that Medicare Part D was associated with significant increases in R&D spending for therapeutic classes that held higher market shares within Medicare. Across a range of empirical settings, the literature establishes a positive relationship between market size and pharmaceutical innovation.

However, these studies do not separate the institutional channel through which drug demand reaches manufacturers. This distinction is particularly important, since the government acts as both a regulator and a key buyer with monopsony power, so policies altering demand may also have implications for the conditions of manufacturer sales

(Lakdawalla, 2018). For example, the expansion of Medicare Part D expanded demand through private prescription drug plans rather than through a rebate and public procurement system. More recent work suggests that this distinction matters. Dranove et al. (2022) exploit the Medicare Part D expansion and find that demand-pull effects are concentrated among non-novel follow-on drugs, with little response among the most scientifically novel candidates, which may reflect the nature of the demand shock, as it was concentrated among elderly patients for whom follow-on treatments are prevalent. More broadly, there is evidence that reductions in access to insured demand inhibit upstream R&D (Agha et al., 2022).

This thesis contributes to this literature by extending the analysis to the Medicaid setting where the market size shock entered through public insurance, and finds an opposite effect in patenting to what is documented within the literature. To interpret this divergence from the literature, this thesis develops a theoretical model to characterise the conditions under which the classical positive market-size prediction reverses, showing that when a monopsony buyer is responsible for demand, the procurement wedge can dominate the demand effect and reduce innovation incentives.

3.2 Medicaid Expansion and Pharmaceutical Innovation

The most closely related study to this thesis is Garthwaite et al. (2021), which examines the impact of the ACA Medicaid expansion on clinical trial activity. Using disease condition Medicaid market shares as a proxy for Medicaid exposure, they find no evidence of a significant innovation response through to 2016. They propose that this null result may reflect the lower expected returns associated with Medicaid relative to reforms such as Medicare Part D, due to manufacturer revenue being constrained by the rebates and price controls embedded within the programme.

While this finding is an important starting point for addressing the question, their analysis leaves several gaps which this thesis addresses. First, the clinical trial outcome observed over a short post-expansion window. Garthwaite et al. investigated clinical trial activity through to 2016, so their estimates only capture the first two years of the Medicaid demand shock. This is short relative to the development horizon for pharmaceutical drug development, which can span between five and twenty years (Brown et al., 2022). As a result, the absence of a response in clinical trials cannot distinguish between an innovation effect that is genuinely absent and one that has affected early-stage R&D decisions but has not reached the clinical trials stage.

Second, the absence of an aggregate clinical-trial response does not confirm whether the innovation effect differs across therapeutic classes. Sutton (1998) argues that pharmaceutical R&D is organised around narrow chemically related drug groups rather than broad therapeutic categories, and finds limited persistence in firms' major follow-on successes within these groups. This suggests that therapeutic-class-level aggregates may conceal meaningful heterogeneity in how firms adjust innovation activity. Instead, an insignificant aggregate relationship may be driven by heterogeneous responses across drug classes. Positive R&D responses may have occurred in classes where the market-size channel dominated, but the aggregate effect may have been offset by negative clinical-trial responses in areas where the expansion weakened innovation incentives.

Third, while Garthwaite et al. propose that the lower returns to the Medicaid market may explain the contrast with earlier market-size studies that find positive innovation responses, this explanation lies outside of their empirical design. Specifically, their analysis does not control for the procurement institutions that may affect manufacturer profits, including rebate pressures and formulary placement on Medicaid's PDL.

This thesis addresses these gaps by extending the analysis across the innovation pipeline, across time, and across therapeutic markets. First, this thesis constructs novel panels linking firms to their drugs in Medicaid's State Drug Utilisation Data, and then to their patenting and FDA approval activity. These innovation outcomes bracket the clinical-trial stage examined by Garthwaite et al., allowing earlier and later innovation responses to be considered separately. Second, the impact of the reform is investigated over a substantially longer post-expansion period, with outcome data extending to 2025, whereas Garthwaite et al. used data ending in 2016. This extension allows delayed innovation responses to be captured more clearly, which is particularly important given the long development horizon in pharmaceutical R&D. Third, class-level heterogeneous responses are examined to determine whether the expansion generated different innovation incentives across the therapeutic markets in which firms operate in, rather than restricting the analysis to an average therapeutic-area effect.

Garthwaite et al. cite the Medicaid procurement mechanism as a potential reason for the inconclusive results; thus this thesis engages directly with this explanation in the empirical analysis with the use of firm-class exposure to Medicaid's PDL. This is complemented by a counterfactual analysis of the clinical and pricing stages of preferred placement on the PDL, alongside a theoretical model in which market expansion interacts with rebates and placement on Medicaid's formulary to shape innovation incentives.

3.3 Medicaid Pricing and Manufacturer Incentives

Medicaid prescriptions are subject to statutory rebates under the MDRP, and each state can negotiate supplemental rebates with manufacturers in exchange for preferred placement on the PDL. Duggan and Scott Morton (2006b) show that these rules create incentives for manufacturers to raise prices for non-Medicaid consumers, since drug prices and reimbursement offered to Medicaid are linked to the prices charged in the private market. Consequently, the results highlight that manufacturers respond differently to demand from Medicaid and private markets, and the ACA may have intensified this distortion by raising the statutory rebate from 15.1% to 23.1% of AMP. Feng et al. (2023) find that the higher statutory rebates relaxed the constraints imposed by Medicaid's most-favoured-customer clause allowing private customers to benefit from lower prices and lower drug spending by 2.5%. Together, these findings show that Medicaid's rebate rules have spillover effects on pricing incentives outside of Medicaid, but the direction of the effect depends on the institutional channel through which the rebate structure operates.

Related structural work shows that pharmaceutical pricing rules can alter manufacturers' pricing behaviour to offset reductions in per-prescription revenues. Dubois et al. (2022) show that firms adjust pricing and bargaining behaviour when price regulations in one market can impact profits elsewhere, indicating that the effect of pricing institutions on manufacturer returns depends on how firms respond strategically across markets.

However, Inderst and Wey (2007) show that larger buyers can raise incentives for incremental forms of product innovation, while weakening incentives for more substantial novel R&D. This distinction is crucial in the Medicaid setting, where the expansion increases market size while also raising exposure to formulary selection, which may weaken incentives for more inventive forms of innovation. This possibility is examined empirically in the Medicaid setting by studying patent applications, which capture early-stage R&D behaviour undertaken in anticipation of future returns. FDA approvals are used to assess whether any upstream innovation response is eventually reflected in regulatory outcomes.

This thesis adds to this literature by extending the analysis to the ACA Medicaid setting, where the pricing channel is assessed empirically at the firm-therapeutic-class level and illustrated using a model of innovation effort subject to PDL placement.

3.4 Innovation Pipeline Timing

The structure and timing of pharmaceutical development are crucial for interpreting different measures of innovation. Pharmaceutical R&D begins with early-stage molecular search and progresses through to clinical trials and eventually reaches the final stage of regulatory approval (Sutton, 1998). Firms typically apply for patent protection well before a candidate drug enters clinical trials (Budish et al., 2015), and clinical development can span between five and twenty years (Brown et al., 2022). Consequently, changes in early-stage innovation measures may not be reflected in downstream measures such as regulatory approvals within a given time period. The empirical results of this thesis provide further evidence of this lagged pipeline structure, where patenting responds to the Medicaid reform while FDA approval responses remain inconclusive within the study window.

4 Data

4.1 Overview

The empirical analysis draws on five sources: the Medicaid State Drug Utilisation Data (SDUD); the Drugs@FDA database; patent bulk data files distributed by the USPTO; the Texas Vendor Drug Program formulary; and the National Library of Medicine's RxNorm and RxClass systems. Table 1 summarises the contribution of each source to the final dataset.

Table 1: Data sources and their roles in the analysis

Source	Identifier	Role
SDUD	NDC, state, year, quarter	Medicaid and non-Medicaid prescription volumes and reimbursement
Drugs@FDA	Application number, sponsor name	FDA drug approval outcomes
USPTO PatentsView	Patent ID, assignee name	Patent counts by therapeutic class
Texas Vendor Drug Program	NDC, effective date	PDL status and prior authorisation indicators
RxNorm / RxClass	RxCUI, ATC code	Therapeutic classification of NDC and active ingredients

The final constructed dataset is a balanced panel at the firm $\times$ ATC Level 1 $\times$ year level, spanning from 2005 to 2025. The empirical analysis uses observations from 2010 onwards, providing a four year pre-expansion period, while avoiding reliance on prescription patterns that are distant from the 2014 reform. All datasets are harmonised to the Level 1 Anatomical Therapeutic Chemical (ATC) classification system, which is described in Section 4.6. Table 2 reports descriptive statistics for each of the four datasets used in the analysis.

It is important to recognise two data limitations. First, the PDL analysis relies primarily on Texas, since it provides historical records in a machine-readable format. Second, most patent assignees are not matched to firms within the Medicaid prescription universe, since many do not have a direct commercial presence. Future work could extend the PDL data to multiple states and draw on proprietary firm linkage data to improve the firm-patent-assignee mapping.

4.2 Medicaid State Drug Utilisation Data

The main source for Medicaid prescription demand is the State Drug Utilisation Data (SDUD), which is maintained by the Centers for Medicare and Medicaid Services (CMS). The SDUD was established alongside the Medicaid Drug Rebate Program and records outpatient prescription drugs reimbursed by state Medicaid programmes at the drug $\times$ state $\times$ quarter level. It covers all fifty states, the District of Columbia and Puerto Rico. Each observation is identified by an 11-digit National Drug Code (NDC) and reports prescription counts, units reimbursed, and reimbursement amounts split into Medicaid

Table 2: Source-level descriptive statistics

Variable	N	Mean	SD	Min	Max
State Drug Utilisation Data					
Unique labeler codes	1,312	-	-	-	-
Unique NDCs	94,128	-	-	-	-
States	52	-	-	-	-
Prescriptions	42,031,131	1,025.41	17,933.34	11	2,519,197.00
Medicaid reimbursement ($000s)	42,031,131	50.11	1,029.85	0	1,061,049.01
Non-Medicaid reimbursement ($000s)	42,031,131	2.32	67.30	0	55,752.11
USPTO PatentsView					
Unique assignees	27,883	-	-	-	-
Unique ATC classes	14	-	-	-	-
Patents per firm-year-ATC	257,484	4.21	11.02	1	540.00
Drugs@FDA					
Unique sponsors	1,465	-	-	-	-
Unique ATC classes	14	-	-	-	-
Approvals per sponsor-year-ATC	14,055	7.68	9.46	1	180.00
Texas Vendor Drug Program					
Unique NDCs	25,728	-	-	-	-
Preferred (PDL)	25,728	0.54	0.50	0	1.00
PT stage pass	25,728	0.68	0.47	0	1.00
Price stage pass	25,728	0.75	0.43	0	1.00

and non-Medicaid components. Annual files between 2005 and 2024 are downloaded via the Medicaid Data portal API, yielding a raw dataset of 42,031,131 observations.

The SDUD plays two roles in the analysis. First, it is used to construct Medicaid exposure measures at the therapeutic-class, which forms the main treatment variable for the analysis. Secondly, once merged with PDL information from the Texas Vendor Drug Program, the SDUD is used to construct exposure measures to formulary placement and pricing pressure at the firm-level.

Table 3: Key variables in the SDUD

Variable	Description
NDC	11-digit code identifying each drug product, split into a 5-digit labeler code, 4-digit product code, and 2-digit package code
Drug name	Proprietary or generic name of the drug
Units reimbursed	Number of units reimbursed by Medicaid in the state-year-quarter cell
Number of prescriptions	Number of prescriptions filled for a given drug in the state-year-quarter cell
Reimbursement amount	Total reimbursement for a given drug, split into Medicaid and non-Medicaid components

4.2.1 National Drug Code

The NDC is an 11-digit product identifier - this contains a 5-digit labeler code, a 4-digit product code, and a 2-digit package code. The SDUD stores the NDC as an 11-digit string without hyphens. However, some datasets remove hyphens or omit leading zeroes. Consequently, the SDUD format is used to normalise all NDCs to ensure that NDC-based joins do not introduce incorrect matches stemming from formatting differences.

4.2.2 Firm identification

The SDUD does not record firm names, so firms are instead identified through their NDC labeler code. However, pharmaceutical firms commonly operate with the use of multiple labeler codes due to subsidiaries or acquisition arrangements. As a result, labeler codes alone cannot be treated as a unique firm identifier in this analysis.

A firm mapping is constructed from the FDA NDC directory. Each labeler code is initially linked to its registered labeler name from the FDA's product and package text files. These names are then clustered into firm-level groups using a text similarity procedure. Labeler names are standardised by converting to lowercase, removing punctuation, and stripping trailing whitespace. Unigram tokenisation is then used to construct a term frequency-inverse document frequency (TF-IDF) representation, and pairwise cosine similarity is computed across all cleaned labeler names. Pairs with a cosine similarity score above 0.85 are treated as matches and are grouped into firm clusters, so that labeler codes with sufficiently similar names are assigned to the same firm. Each cluster is assigned the name of its first member as the canonical firm identifier. Merging the resulting firm mapping into the SDUD by labeler code allows for 40,003,656 of 42,031,131 records to match, which corresponds to a match rate of 95.18%.

4.2.3 Therapeutic classification

Each NDC in the SDUD is mapped to an ATC Level 1 class using the RxNorm API. First, each NDC is submitted to the RxNorm NDC status endpoint in order to obtain its RxCUI identifier. This is then used to retrieve ATC codes through the RxNorm ATC lookup endpoint. ATC Level 1 is defined by the first character of the ATC code. This procedure classifies 96.10% of SDUD records. Records that cannot be classified are excluded from the estimation sample. In cases where a single NDC maps to more than one ATC class, the record is duplicated so that each NDC-class combination appears separately. These duplicated records are then aggregated within class cells in the final panel.

4.3 Texas Preferred Drug List Data

PDL data is taken from the Texas Vendor Drug Program, which publishes machine-readable historical records of Medicaid drug coverage decisions. Texas is selected because it is one of the few states to make its complete formulary history available in a machine-readable format with drug-level effective and end dates. Unlike many other states, whose PDL data are available only as PDFs or current cross-sections, the Texas data allow a time-varying measure of drug-level preferred status to be constructed. The formulary file records the following fields for each listed drug: the NDC, the Medicaid coverage start and end dates, a coverage status indicator, a flag indicating whether PDL-related prior authorisation is required, and a flag indicating whether clinical prior authorisation is required.

Two variables are constructed to separate the clinical and pricing stages of PDL selection. PT_Stage_Pass captures drugs that clear the Pharmacy and Therapeutics clinical review, i.e. these are drugs which are clinically acceptable. Price_Stage_Pass captures drugs that clear the pricing stage, meaning that their net cost after rebates is sufficiently competitive for preferred placement. A drug is therefore classified as being on the PDL when it is covered by Medicaid and clears both the clinical and pricing stages.

4.3.1 Effective date treatment

A drug is classed as having active coverage in a given year-quarter if its coverage start date falls prior to the end of the quarter and it either has no coverage end date or the coverage end date is after the start of the quarter. Drugs for which no end date is present, or for which the end date is in 2025 or later, are treated as currently active.

4.3.2 Linking to the SDUD

The Texas PDL is merged with the SDUD at the full 11-digit NDC level - it is matched on the labeller, product, and package codes simultaneously. NDC records without a corresponding Texas formulary entry receive missing values for their PDL indicators as opposed to being excluded. Missing PDL values are subsequently set to zero when constructing the firm-level PDL exposure measures - absence from the Texas Drug Vendor Program record likely reflects a drug that has not obtained preferred status rather than a data gap.

4.3.3 Representativeness

Since the PDL analysis relies on a single state's formulary, the extent to which the Texas PDL is representative of the Medicaid PDL choices in other states is assessed by computing pairwise overlap rates with comparison states. This representativeness analysis is limited by data availability - the majority of states do not publish machine-readable PDL records with historical coverage dates, so the comparison can only be conducted for the subset of states whose formulary data are available in a structured format. On this basis, New York, California, and Washington are used as comparison states. Overlap is measured in both directions as the share of drugs on one state's PDL that also appear on the other state's PDL, matched at the labeller and product level of the NDC.

The results indicate that 95.0% of drugs on the Texas PDL also appear on the New York PDL, and 81.2% also appear on the Washington PDL. Overlap with California is substantially lower at around 20% in both directions, reflecting California's distinct formulary structure. The overlap analysis suggests that the Texas PDL shares are somewhat representative of the preferred drug lists of other large states. However, it should not be treated as a direct proxy for every state's formulary decisions, especially given the weaker overlap with California. In this thesis, the Texas PDL is used primarily to capture the two-stage clinical-and-price structure of PDL selection, rather than to measure the exact set of preferred drugs nationwide.

Table 4: Pairwise overlap of state Preferred Drug Lists

State 1	State 2	Share of State 1 PDL also on State 2 PDL	Share of State 2 PDL also on State 1 PDL
New York	Texas	0.375	0.950
California	Texas	0.196	0.197
Washington	Texas	0.223	0.812
California	New York	0.722	0.230
California	Washington	0.384	0.106

This creates measurement error when PDL exposure in Texas is used as a proxy for firm's exposure to Medicaid across the United States. Since this measurement weakens the link between the constructed PDL variable and a firm's true national exposure, the estimated innovation response is likely to be attenuated towards zero. Therefore regression estimates using PDL exposure should be interpreted as a lower bound on the effect of formulary placement.

Future work could extend this approach by obtaining PDL records from additional states and constructing state-specific firm-class PDL exposure measures. These state-level exposure measures could then be averaged across states to obtain a multi-state measure of firms' exposure to PDL placement. As more state formularies are added, this measure would better approximate a firm's overall exposure to the PDL stages.

4.4 FDA Approval Data

Approval data for drugs is obtained from the Drugs@FDA database, which records all drug applications submitted to the FDA. The applications, submissions and products files identify sponsor names, application numbers, approval status and the associated active ingredients for approved drugs. The sample is restricted to approvals occurring on or after 1 January 2005 to align with the SDUD coverage window.

4.4.1 Therapeutic classification

Therapeutic classification of FDA-approved drugs occurs at the ingredient level. To do this, active ingredients strings are standardised and split to account for multi-ingredient products. Each combination of normalised ingredients are then queried against the NLM's RxClass API. This maps each ingredient to one or more ATC codes. Ingredient-level results are collapsed at the application level so that each application is linked to the union of ATC Level 1 classes associated with its active ingredients. Applications for which no ATC class can be assigned are excluded. This procedure classifies 48,143 of 50,368 application-product observations giving a match rate of 95.58%.

In cases where an application maps to multiple ATC Level 1 classes, the application is counted once within each applicable class at the aggregation stage. This is analogous to the treatment of multi-class NDCs in the SDUD.

4.4.2 Firm-NDC matching

Since the Drugs@FDA database identifies firms by their name as opposed to their NDC labeler code, an additional matching step is required. FDA sponsor names and NDC labeler names are standardised by: conversion to uppercase, removal of non-alphanumeric characters, and stripping of common corporate suffixes such as Inc, Corp, LLC. A three-stage merging procedure is then applied. In the first stage, firms are matched exactly by their cleaned name. In the second stage, unmatched FDA sponsors are linked to NDC labeler names via a root-token match, where the root is the first alphanumeric token which is at least four characters long. Finally, remaining unmatched firms are linked using Jaro-Winkler string similarity at a threshold of 0.90, with the highest-scoring NDC match being used. Under this process, a match rate of 82.26% is achieved.

4.5 Patent Data

Patent data is obtained from the USPTO PatentsView data files. PatentsView harmonises patent records and provides disambiguated assignee information. Specifically, for this study, five files are used: the base patent file, which provides grant dates; the current CPC classification file; the disambiguated assignee file; the CPC group title file; and the WIPO technology field classification file.

4.5.1 Relevant patents

The universe of patents is filtered in two steps. First, patents are restricted to those classified under the WIPO technology field "Pharmaceuticals". Within this set, patents are restricted to those that fall under the "A61P" subclass - this allows for the study to focus on patents with a stated therapeutic use. Patents classified under A61P but not assigned to the WIPO Pharmaceuticals field are also included, as some patent-level WIPO assignments may be incomplete. This allows for the study to focus on patents that are most likely to be related to therapeutic drug innovation.

It is important to note that the CPC classifications are not perfect measures of pharmaceutical invention of a patent, since CPC classes are assigned during the patent examination process. Consequently, the A61P restriction may exclude some relevant patents relating to delivery or formulation, while also retaining some patents that are classified as pharmaceutical but less directly related to new therapeutic agents.

4.5.2 Therapeutic classification

The A61P subclass is divided into CPC groups corresponding to different therapeutic uses. A manually constructed crosswalk is used to map each A61P group to an ATC Level 1 class, with the crosswalk following the structure of the USPTO's A61P classification. Selected entries from this crosswalk are shown in Table 5. For example, patents related to cardiovascular activity are classified under A61P9, so these are mapped to ATC class C, while patents related to nervous system activity fall under A61P25, thus they are mapped to ATC class N. In cases where a patent falls under multiple different A61P CPC groups, it is counted once within each applicable class.

Table 5: CPC A61P to ATC Level 1 crosswalk (selected entries)

CPC Group	Therapeutic area	ATC Level 1
A61P1	Gastrointestinal	A
A61P7	Blood	B
A61P9	Cardiovascular	C
A61P17	Dermatology	D
A61P15	Genito-urinary and sex hormones	G
A61P5	Endocrine	H
A61P31	Anti-infectives	J
A61P35	Antineoplastics	L
A61P19	Musculo-skeletal	M
A61P25	Nervous system	N
A61P33	Antiparasitic	P
A61P11	Respiratory	R
A61P27	Sensory organs	S

4.5.3 Firm-NDC matching

Patent assignee names are matched to the NDC labeler firm clusters using the same three-stage procedure applied to sponsors in the Drugs@FDA dataset. Cleaned assignee names are first matched to labeler names exactly. The remaining unmatched assignees undergo root-token matching followed by a Jaro-Winkler fuzzy match at a threshold of 0.90.

Out of 28,143 unique assignees, 7,566 were matched to the NDC firm universe, which corresponds to a match rate of 27.1%, but this understates patent-level coverage, where 46.8% of patents were matched to a firm. To assess whether the matched and unmatched patent samples differ systematically, Tables 13 and 14 compare the therapeutic and temporal composition of the matches. The distribution of patents across ATC Level 1 is similar across the two groups, and the shares of patents before and after 2014 are similar between the matched and unmatched group. This provides support for the patent-firm matching error being non-systematic.

4.6 ATC Level 1 Classes

Ultimately, all prescription, FDA approval, and patent records are aggregated to ATC Level 1, thus Table 6 reports the therapeutic classes under each ATC level. The source-specific mapping procedures have been described in the relevant subsections above.

4.7 Firm Harmonisation

The main datasets identify firms using different methods. For example, the SDUD uses NDC labeler codes, which are first grouped into firm clusters and then used as the firm universe for the analysis. The Drugs@FDA instead uses sponsor names to identify firms, while PatentsView identifies firms using their disambiguated assignee names. FDA sponsors and patent assignees are therefore linked back to NDC firm clusters. Table 7 summarises the match rates across the main linking steps.

Table 6: ATC Level 1 therapeutic classification

Code	Therapeutic area
A	Alimentary tract and metabolism
B	Blood and blood-forming organs
C	Cardiovascular system
D	Dermatologicals
G	Genito-urinary system and sex hormones
H	Systemic hormonal preparations
J	Anti-infectives for systemic use
L	Antineoplastic and immunomodulating agents
M	Musculo-skeletal system
N	Nervous system
P	Antiparasitic products, insecticides and repellents
R	Respiratory system
S	Sensory organs
V	Various

Table 7: Match rates across data linking steps

Linking step	N	Match rate
SDUD labeler codes to firm clusters	42,031,131	95.18%
SDUD NDC to RxNorm ATC	42,031,131	96.10%
Drugs@FDA applications to ATC	50,368	95.58%
Drugs@FDA sponsors to NDC firms	1,465	82.26%
Patent assignees to NDC firms	28,143	27.10%

4.8 Panel Construction

The final panels are constructed separately for the patent and FDA approval outcomes. The firm-by-class structure is designed to recover variation in innovation incentives within therapeutic areas, consistent with Sutton's (1998) account of pharmaceutical R&D as heterogeneous across product areas. Both patent counts and FDA approvals are collapsed to the firm $\times$ ATC Level 1 $\times$ year level. The SDUD prescription and reimbursement variables are first aggregated to the firm $\times$ ATC Level 1 $\times$ year level and are then merged with the innovation panels. PDL variables from the Texas Vendor Drug Program are merged at the firm $\times$ ATC Level 1 level - details relating to their exact construction are described in the Methodology section.

Each panel is balanced over the firm-class combinations observed in the data. In cases where no patent or approval observations are recorded for a given firm-class-year, an outcome value of zero is assigned. Summary statistics for each of the balanced panels are reported in Table 8. Histograms for variation in constructed PDL exposure variable in the interval $(0, 1]$ can be found in Figure 9.

Table 8: Balanced panel summary statistics

Variable	N	Mean	SD	Min	Max
SDUD Patents panel					
Unique firms	23,580	-	-	-	-
Unique ATC classes	14	-	-	-	-
Years covered	21	-	-	-	-
Patents	6,932,520	0.16	2.27	0	540.00
PDL exposure	6,932,520	0.00	0.02	0	1.00
Ex Ante Medicaid Demand	6,932,520	0.07	0.09	0	0.35
Ex Post Medicaid Demand	6,932,520	0.07	0.08	0	0.33
SDUD FDA approvals panel					
Unique firms	675	-	-	-	-
Unique ATC classes	14	-	-	-	-
Years covered	21	-	-	-	-
FDA approvals	198,450	0.54	4.50	0	235.00
PDL exposure	198,450	0.01	0.08	0	1.00
Ex Ante Medicaid Demand	198,450	0.07	0.10	0	0.36
Ex Post Medicaid Demand	198,450	0.07	0.08	0	0.26

5 Methodology

5.1 Empirical Strategy

The empirical analysis proceeds in three stages. First, an event-study specification is utilised to estimate the aggregate innovation response to Medicaid exposure across therapeutic classes. Second, PDL exposure is incorporated to examine whether the innovation response differs depending on firms' exposure to Medicaid procurement mechanisms. Third, the post-expansion innovation response is allowed to vary for each therapeutic area in order to assess heterogeneity in innovation responses. The main outcomes are patent counts and FDA approvals, measured at the firm-therapeutic-class level.

The specifications are motivated by the market-size channel in pharmaceutical innovation, where an increase in expected demand should increase the expected return to R&D and therefore raise innovation incentives (Dubois et al., 2015; Finkelstein, 2004). Since the additional demand flows through a public buyer with rebate authority and formulary control, the Medicaid setting complicates this result. For this reason, the empirical analysis first aims to specifically examine whether the innovation response differs with a firm's exposure to the PDL.

5.2 Identification

The empirical strategy treats the ACA Medicaid expansion as a national shock to the expected profitability of drugs in therapeutic classes with different levels of pre-existing Medicaid exposure. While the expansion itself was adopted at the state level, innovation in the pharmaceutical industry is not a state-specific decision. Pharmaceutical firms develop drugs for the national markets, and as a result, the expected returns depend on anticipated sales across the United States as opposed to the realised expansion within any single expansion state. For this reason, the empirical analysis does not leverage the staggered state adoption policy as the main source of variation.

Garthwaite et al. (2021) studies anticipatory R&D responses to Medicaid, where the relevant shock is the announcement of the ACA in 2010. However, a 2012 Supreme Court decision made the expansion of Medicaid voluntary at the state level, which introduced uncertainty over the magnitude of the shift in market size. Consequently, this thesis examines the innovation effects following 2014, as this reflects the time when states could expand Medicaid coverage.

The expansion increased the number of individuals covered by Medicaid, and therefore increased the volume of prescriptions to be reimbursed through Medicaid. Since some therapeutic areas in Medicaid accounted for a larger share of prescriptions prior to the expansion, these areas were more exposed to the market-size shift compared with classes in which Medicaid played a smaller role. This indicates the shock did not hit all therapeutic classes equally. Therefore, the identifying variation stems from cross-class differences in pre-expansion Medicaid exposure. Firms operating in high-exposure classes are interpreted as facing a larger Medicaid demand shock after 2014. If the expansion increased expected demand in high-Medicaid classes, standard market-size models would predict a stronger innovation response in those classes after 2014 relative to less-exposed classes.

The main identifying assumption for each of the research questions is parallel trends. That is, absent the ACA Medicaid expansion, innovation outcomes in high and low

Medicaid-exposure classes would have followed a similar trajectory over time.

This is plausible since the constructed Medicaid exposure measure is largely determined by the patient population using drugs in each therapeutic class before 2014. Some classes have higher Medicaid utilisation due to treating conditions that are more prevalent amongst lower income Medicaid-eligible populations. This does not necessarily imply that those classes were on different innovation trajectories before the reform. Instead, innovation trends are likelier to depend on factors such as scientific opportunity. Consequently, there is no reason to expect classes with different exposure levels to Medicaid to have innovation outcome divergence absent the ACA expansion.

Since pharmaceutical R&D operates over long development cycles, patents and FDA approvals observed after 2014 may reflect investment decisions initiated prior to expansion implementation. The event-study specification can identify differential pre-expansion trends and the timing of any post-expansion divergence. Small and statistically insignificant pre-2014 coefficients would support the parallel trends assumption, indicating that innovation followed similar trajectories amongst classes with differing Medicaid exposures prior to the reform.

Identification also requires that there were no contemporaneous shocks around 2014 that differentially affected innovation in high-Medicaid-exposure classes. Year fixed effects are used to absorb shocks to the pharmaceutical industry such as FDA regulation changes or industry-wide shifts in R&D that affect all therapeutic areas within a given year. Given the fixed-effects, a confounding shock would therefore need to occur around 2014 and affect some therapeutic classes more than others. There are no clear candidates for such a shock, and other ACA provisions, including changes to private insurance markets and coverage mandates, affected the pharmaceutical sector more broadly and their effect would be absorbed by year fixed effects.

The specifications include firm-class fixed effects and year fixed effects. Firm-class fixed effects absorb time-invariant differences in innovation levels across firm-class pairs. This is important since firms may specialise in particular therapeutic areas - separate firm and class fixed effects may not be able to capture this source of heterogeneity. Year fixed effects absorb aggregate shocks to pharmaceutical innovation that are common across classes in a given year. Standard errors are clustered at the firm level to account for serial correlation in innovation outcomes within firms over time, since a firm's current innovation activity may influence its future innovation behaviour.

5.3 Construction of Key Variables

5.3.1 Ex-Ante Medicaid Market Share

Following the empirical framework established by Garthwaite et al. (2021) in their study of Medicaid expansions, the main exposure measure in this paper is based on ex-ante Medicaid market shares, calculated using prescription shares prior to the ACA Medicaid expansion. For each therapeutic class c, this predetermined variable is calculated as:

$$\text{MedicaidShare}_c = \frac{\sum_{t=2010}^{2013} \text{MedicaidPrescriptions}_{c,t}}{\sum_{t=2010}^{2013} \text{TotalPrescriptions}_{c,t}} \tag{1}$$

Classes with higher pre-expansion Medicaid shares are interpreted as those more exposed to the demand consequences of expansion and ex-ante construction avoids using post-treatment outcomes to define treatment intensity. Therapeutic classes may differ systematically in patient demographics or product characteristics, but these differences are likely to be persistent over time, thus the firm-class fixed-effects structure absorbs such time-invariant sources of heterogeneity.

5.3.2 Ex-Post Medicaid Market Share

As a complement to the ex-ante measure, an ex-post exposure proxy based on realised demand shifts in the early years of the reform is also calculated. For each therapeutic class, this is constructed as:

$$\text{MedicaidShare}_c = \frac{\sum_{t=2014}^{2015} \text{MedicaidPrescriptions}_{c,t}}{\sum_{t=2014}^{2015} \text{TotalPrescriptions}_{c,t}} \tag{2}$$

This captures the extent to which a therapeutic class actually experienced increased Medicaid demand in the immediate post-expansion period. Since this measure is closer to the realised response of the market, it may reflect the environment faced by firms in the Medicaid market after expansion. Given that this measure is not predetermined, estimates using this variable should be interpreted as a robustness check.

5.3.3 PDL Exposure

To examine exposure to the Medicaid pricing mechanisms, a measure of preferred drug list exposure for firm f in class c is constructed using the universe of drugs d over the baseline period:

$$\text{PDL\_Exposure}_{f,c} = \frac{\sum_{t=2010}^{2014} \sum_d \mathbf{1}\{d \in f, c, t\} \cdot \mathbf{1}\{d \in \text{PDL}_t\}}{\sum_{t=2010}^{2014} \sum_d \mathbf{1}\{d \in f, c, t\}} \tag{3}$$

Higher levels of PDL exposure indicate that a larger share of a firm's drug portfolio in a therapeutic class is subject to formulary selection and the pricing environment associated with preferred placement on the PDL. Empirically, this variable is used as a proxy for exposure to procurement institutions rather than as a direct measure of rebates or negotiated prices. While this measure utilises data up to 2014, which is when the ACA expansion occurred, Texas was a non-expansion state, therefore the measure is predetermined with respect to the ACA.

5.3.4 PDL Counterfactual Measures

Two time-invariant firm-class measures are constructed from the Texas Vendor Drug Program formulary data for use in the counterfactual analysis specification. Each is defined as the share of a firm's NDCs in class c satisfying a given PDL criterion, averaged over the baseline period:

$$\text{PT\_Pass}_{f,c} = \frac{\sum_{t=2010}^{2014} \sum_d \mathbf{1}\{d \in f, c, t\} \cdot \mathbf{1}\{\text{PT\_Stage\_Pass}_{d,t} = 1\}}{\sum_{t=2010}^{2014} \sum_d \mathbf{1}\{d \in f, c, t\}} \tag{4}$$

$$\text{Price\_Pass}_{f,c} = \frac{\sum_{t=2010}^{2014} \sum_d \mathbf{1}\{d \in f, c, t\} \cdot \mathbf{1}\{\text{Price\_Stage\_Pass}_{d,t} = 1\}}{\sum_{t=2010}^{2014} \sum_d \mathbf{1}\{d \in f, c, t\}} \tag{5}$$

These proportions are treated as time-invariant firm-class characteristics. Drugs with no match in the Texas formulary are assigned a value of zero for all three measures.

5.4 Research Questions

5.4.1 RQ1: Overall Innovation Response

The first research question asks whether therapeutic classes with greater Medicaid exposure experienced different innovation trajectories following the ACA Medicaid expansion. This is estimated using the following event-study specification:

$$Y_{f,c,t} = \alpha_{f,c} + \lambda_t + \sum_{\tau \neq 2013} \beta_\tau \left(\mathbf{1}\{t = \tau\} \times \text{MedicaidShare}_c \right) + \varepsilon_{f,c,t} \tag{6}$$

where $Y_{f,c,t}$ is an innovation outcome for firm f, therapeutic class c, and year t. The main outcomes are patent counts and FDA approvals. MedicaidShare_c measures pre-expansion Medicaid exposure in class c, $\alpha_{f,c}$ denotes firm-by-class fixed effects, and λ_t denotes year fixed effects. Since the coefficient for 2013 is omitted, each β_τ captures the differential innovation response associated with higher pre-expansion Medicaid exposure in year τ, relative to 2013.

This specification is estimated using both fixed-effects ordinary least squares (FEOLS) and fixed-effects Poisson (FE-Poisson) models. The linear FEOLS model provides a baseline and accommodates the event-study structure directly. By plotting the linear interaction coefficients, the FEOLS specification provides a straightforward, visually intuitive assessment of both the parallel trends assumption and the dynamic evolution of the treatment effect over time.

Innovation outcomes, such as patent grants and FDA approvals, are non-negative, and negatively skewed integer counts. Given this underlying data-generating process, the specification is also estimated using a FE Poisson model. The Poisson estimator is suitable in this setting since it can handle count data and zero-valued observations without requiring additional adjustments in functional form. Moreover, provided the conditional mean is correctly specified, the Poisson estimator is consistent even if the underlying data exhibits significant overdispersion Wooldridge (1999). Therefore, estimating both linear and Poisson specifications allows the analysis to assess whether the main findings are sensitive to the treatment of the count nature of the outcomes.

Both ex-ante and ex-post exposure measures are used to measure MedicaidShare_c. The ex-ante measure is based on pre-2014 Medicaid prescription shares and captures exposure that is predetermined with respect to the policy change. The ex-post measure instead utilises realised post-expansion demand shifts and is interpreted more cautiously, as it potentially contains the contemporaneous market response induced by the ACA.

5.4.2 RQ2: Market Size and PDL Exposure

The second research question asks whether the innovation response to Medicaid expansion differs depending on a firm's exposure to the PDL institutions. The event-study specification takes on form:

$$
\begin{aligned}
Y_{f,c,t} = {} & \alpha_{f,c} + \lambda_t \\
& + \sum_{\tau \neq 2013} \beta_\tau \left(\mathbf{1}\{t = \tau\} \times \text{MedicaidShare}_c\right) \\
& + \sum_{\tau \neq 2013} \gamma_\tau \left(\mathbf{1}\{t = \tau\} \times \text{PDL\_Exposure}_{f,c}\right) + \varepsilon_{f,c,t}
\end{aligned}
\tag{7}
$$

where $\text{PDL\_Exposure}_{f,c}$ measures the extent to which firm f's portfolio in class c is exposed to Medicaid's formulary institutions. This is constructed as the share of the firm's drugs in a given therapeutic class that appear on the PDL. The coefficients on the MedicaidShare_c interactions capture how innovation varies with Medicaid demand exposure in each year relative to 2013. The coefficients on the $\text{PDL\_Exposure}_{f,c}$ interactions capture how innovation varies with firms' exposure to Medicaid formulary institutions over the same period.

While this model does not provide a clean structural decomposition of market size and pricing pressures, PDL exposure is utilised as a proxy rather than a direct measure of rebate extraction or bargaining intensity. Higher PDL exposure being associated with weaker post-expansion innovation responses would be consistent with the institutional setting of Medicaid weakening the standard market-size effect.

5.4.3 RQ3: Therapeutic-Class Heterogeneity

The third research question investigates whether the effect of the Medicaid expansion differs across therapeutic classes. Pharmaceutical R&D is not organised around a single uniform innovation process across the industry. Sutton (1998) suggests that the drug discovery process operates across distinct product areas, with chemically related groups displaying limited persistence in innovation successes. This implies that the aggregate response to the Medicaid expansion recorded in RQ1 and RQ2 may conceal heterogeneous responses across the therapeutic markets. This is particularly relevant within this framework as the underlying market structure may vary across therapeutic areas in ways that are unobserved by the pooled specification.

To examine this, a triple-interaction specification is specified. This allows the post-expansion effect of Medicaid exposure to vary by the ATC Level 1 class:

$$
Y_{f,c,t} = \alpha_f + \delta_c + \lambda_t + \sum_{a \in \mathcal{A}} \theta_a \left(\mathbf{1}\{t \geq 2014\} \times \text{MedicaidShare}_c \times \mathbf{1}\{c = a\}\right) + \varepsilon_{f,c,t} \tag{8}
$$

where a is a specific ATC within the universe of ATC Level 1 therapeutic areas $\mathcal{A}$, and θ_a captures the post-2014 differential response associated with Medicaid exposure in therapeutic area a. ATC class V captures products outside of the more clinically defined ATC categories, and is omitted from the specification to serve as the reference group. Firm and class effects are separated in this specification due to computational constraints associated with the larger number of interaction terms. These estimates provide evidence on the heterogeneity of the response to the ACA and help assess whether the aggregate results are driven by particular therapeutic areas.

5.4.4 PDL Counterfactual Analysis

In order to examine the innovation mechanism more directly, a counterfactual design is estimated by leveraging the two-stage structure of Medicaid Preferred Drug List (PDL) placement. A drug receives preferred status if and only if it clears both a clinical (P&T) stage and a pricing stage. The following specification separately identifies exposure to each stage and passing both stages jointly:

$$\begin{aligned} Y_{f,c,t} = \alpha_{f,c} + \lambda_t &+ \beta_1 \left(\text{PT\_Pass}_{f,c} \times \text{Post}_t\right) \\ &+ \beta_2 \left(\text{Price\_Pass}_{f,c} \times \text{Post}_t\right) \\ &+ \beta_3 \left(\text{PDL\_Pass}_{f,c} \times \text{Post}_t\right) \\ &+ \varepsilon_{f,c,t} \end{aligned} \tag{9}$$

where $\text{PT\_Pass}_{f,c}$, $\text{Price\_Pass}_{f,c}$ are proportions measuring the share of a firm's drugs that pass the clinical stage and the pricing stage for the PDL, $\text{PDL\_Pass}_{f,c}$ measures the share of a firm's drugs securing preferred status by passing both stages.

The coefficients β_1 and β_2 capture the post-expansion change in innovation associated with a higher share of a firm's drugs passing the clinical and price stages, respectively. β_3 captures the additional innovation response associated with a higher share of drugs passing both stages and therefore securing preferred status.

This exercise is not intended to identify the full structural effect of the PDL process. However, it tests whether the innovation response differs across firms depending on their position within the PDL. If firms which pass both stages more often experience more favourable innovation responses, then this would suggest that preferred placement on the PDL affects the profitability of innovation.

6 Empirical Results

6.1 RQ1: Overall Innovation Response

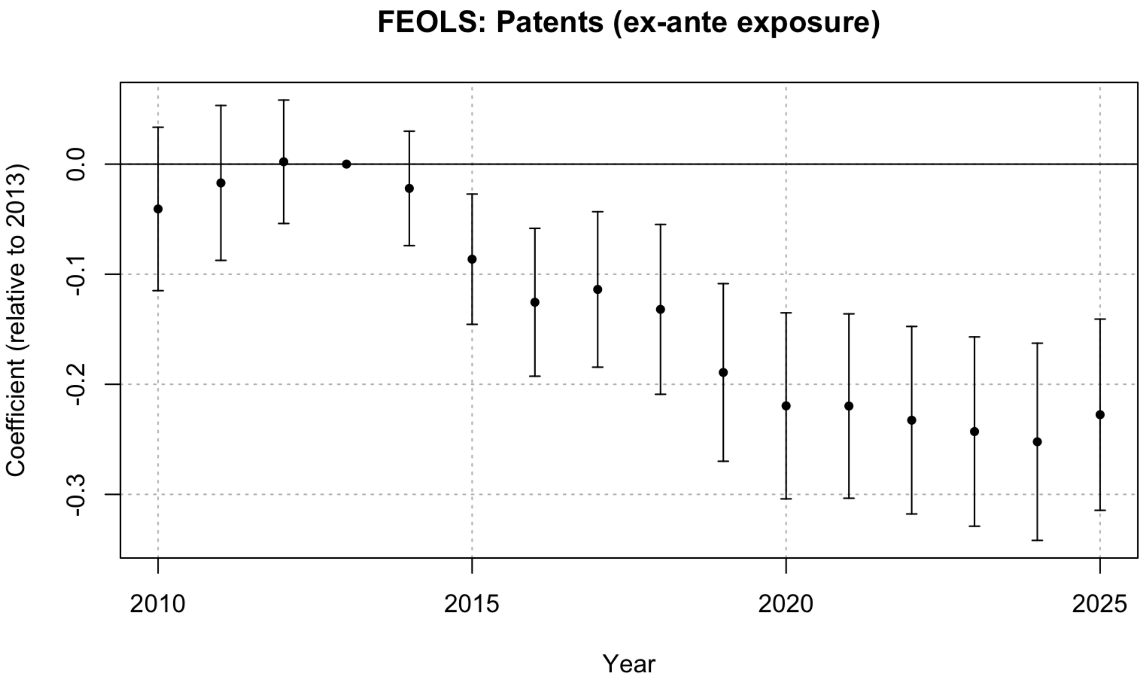

Figure 1: FEOLS event study: patents, ex-ante Medicaid exposure

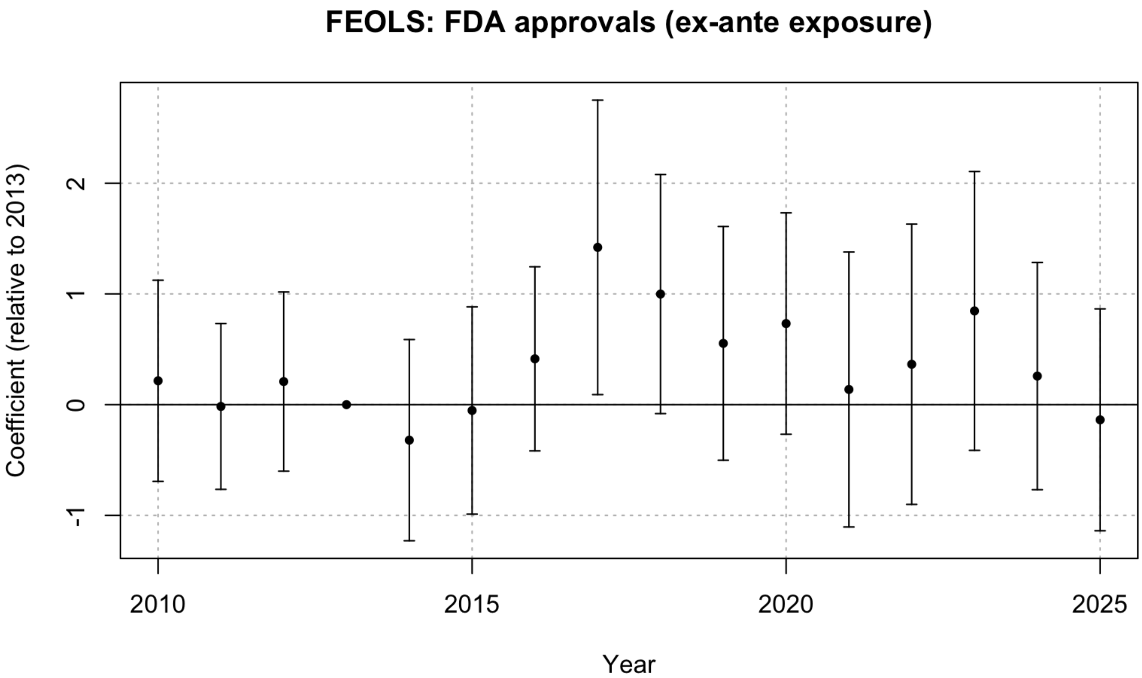

Figure 2: FEOLS event study: FDA approvals, ex-ante Medicaid exposure

The pre-expansion coefficients between 2010 and 2012 are small and statistically insignificant across both specifications, which is consistent with the assumption of parallel trends in firm patenting behaviour between areas with differential Medicaid market shares. Under the FEOLS specification with ex-ante exposure (Table 15), the post-expansion coefficient is insignificant in 2014 and then becomes significant in 2015 at -0.086, before growing in magnitude over time and peaking at -0.252 in 2024. Since the exposure to Medicaid is measured as a therapeutic class's market share within Medicaid, the estimated coefficients correspond to a one-hundred percentage point increase in Medicaid share. Consequently, a therapeutic class with a 10 percentage-point higher Medicaid market share would be associated with 0.025 fewer patents per firm-class-year by 2024. This corresponds to about 16 percent of the sample mean patent count (0.16). Noticeably, the coefficients stabilise between 2020 and 2025 in the range of -0.220 to -0.252, indicating that the negative response persists.

The gradual onset of the effect is expected given that patent outcomes are recorded by the date of award rather than the application date. Thus, patents observed shortly after 2014 may include applications prior to the ACA, so the timing should not be interpreted as the exact timing of underlying R&D decisions. Additionally, the effect also reflects transition dynamics in R&D decisions. Firms would not immediately adjust their innovation decisions in response to policy shifts, and pricing in the new Medicaid market conditions into R&D plans may take time given the long development horizons involved in pharmaceutical innovation. Both of these factors therefore contribute towards the increasing magnitude of the coefficients where stabilisation is seen between 2020 and 2025.

The results from FDA approvals contrast with the patent findings. The post-expansion coefficients are statistically insignificant with wide confidence intervals. Whereas the patent FEOLS estimates grow monotonically from -0.086 in 2015 to -0.252 in 2024, the FDA FEOLS estimates fluctuate between -0.321 and 1.421 without a sustained direction. The isolated significance in 2017 and 2018 is consistent with sampling variation given the number of post-expansion coefficients tested. Overall, the results for FDA approvals provide inconclusive evidence of a response to Medicaid demand exposure at the end stage of the pipeline.

The difference between the patent and FDA results is consistent with the structure of the pharmaceutical R&D pipeline documented in the literature. Sutton (1998) suggests pharmaceutical development follows a staged process, which can span over a decade (Brown et al., 2022). Patenting activity captures an earlier stage of this process relative to FDA approvals, and so changes in patenting appear earlier than approvals. Therefore, the ACA policy shock may plausibly affect patenting without yet generating a response in FDA approvals. Changes in patenting decisions would feed through into drugs under active development, subsequently appear in clinical trial activity, and only reach FDA approvals once the affected drug cohort arrives at the final stage of the development pipeline. Under this interpretation, the insignificant response of FDA approvals reflects that insufficient time has passed for the upstream adjustment to result in detectable changes at the approval stage as opposed to contradicting the patent findings.

This is consistent with Garthwaite et al. (2021), who study the response of clinical trial activity with three years of post-ACA data and find an insignificant innovation response. Clinical trial activity generally occurs later in the development process than early patenting activity, and given that the patent event-study displays a gradual build-up over nearly a decade, a three-year window is likely too short to detect a response even at the clinical trial stage.

6.2 RQ2: Medicaid Demand and PDL Exposure

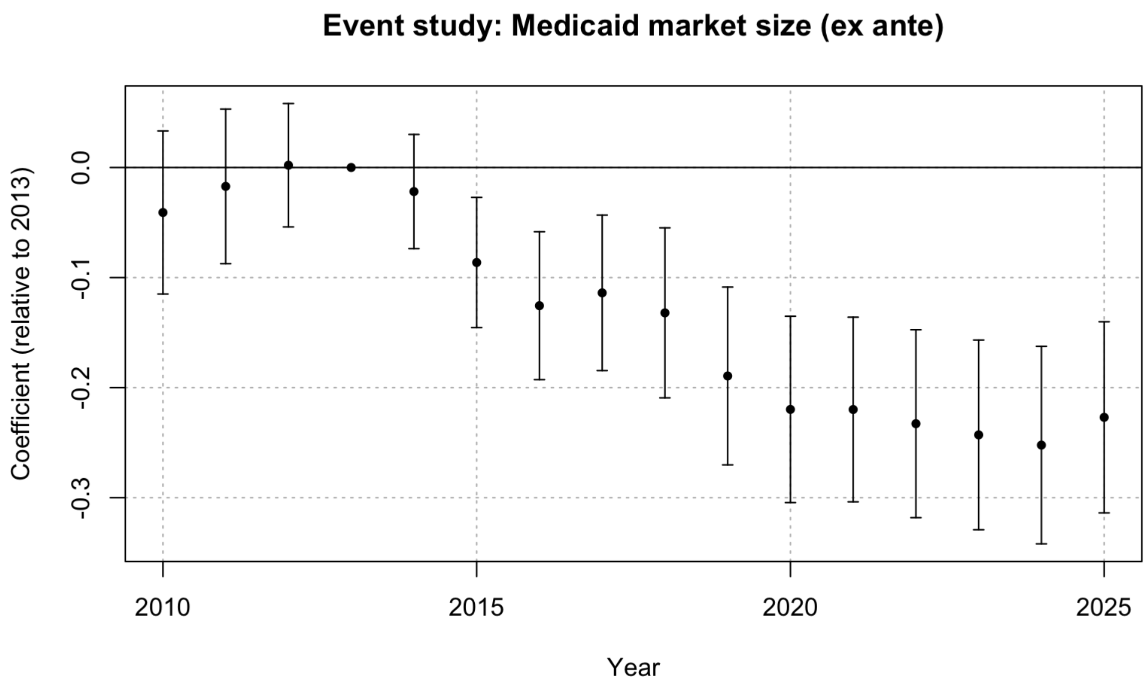

Figure 3: FEOLS event study: patents, Medicaid exposure (ex-ante)

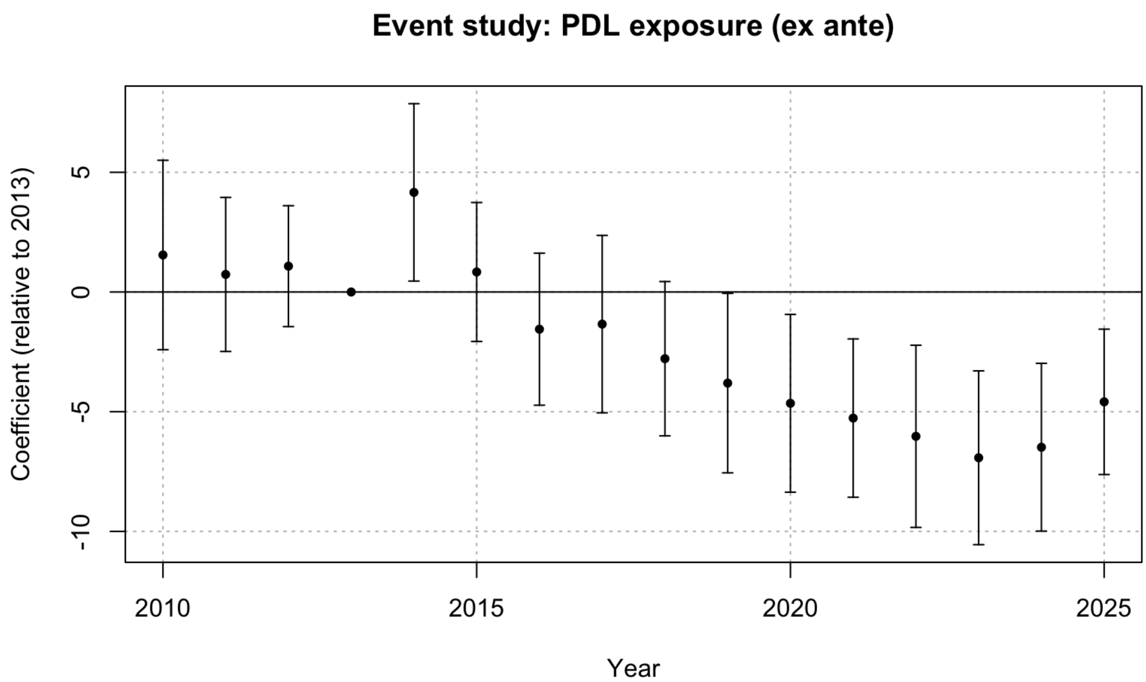

Figure 4: FEOLS event study: patents, PDL exposure (ex-ante)

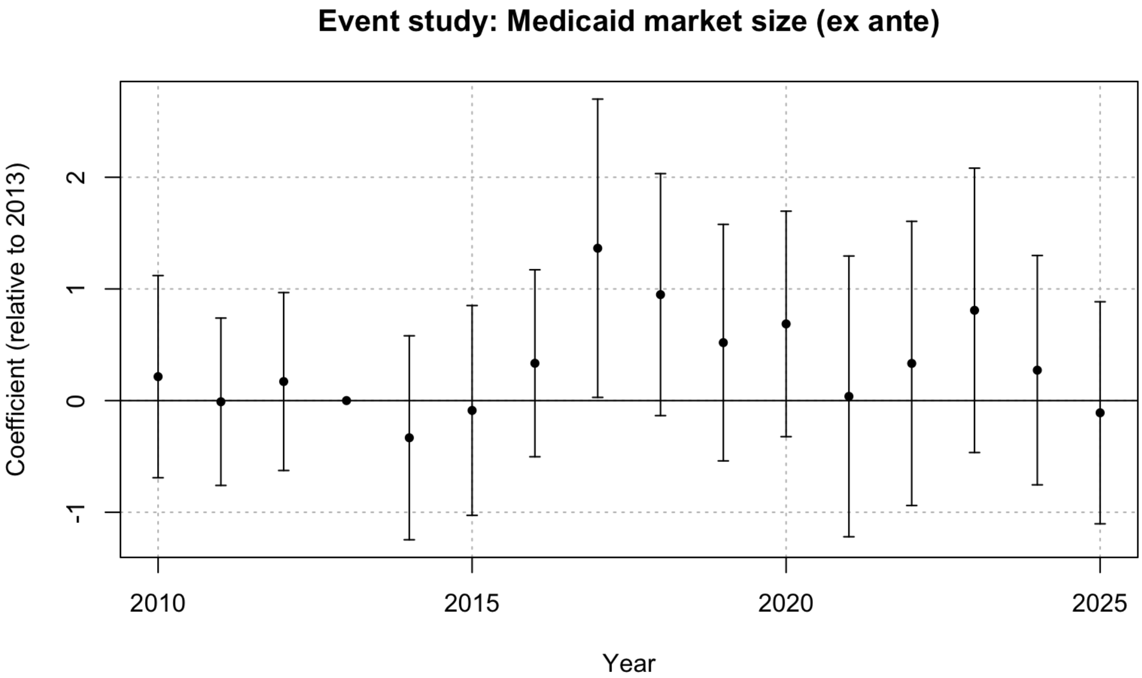

Figure 5: FEOLS event study: FDA approvals, Medicaid exposure (ex-ante)

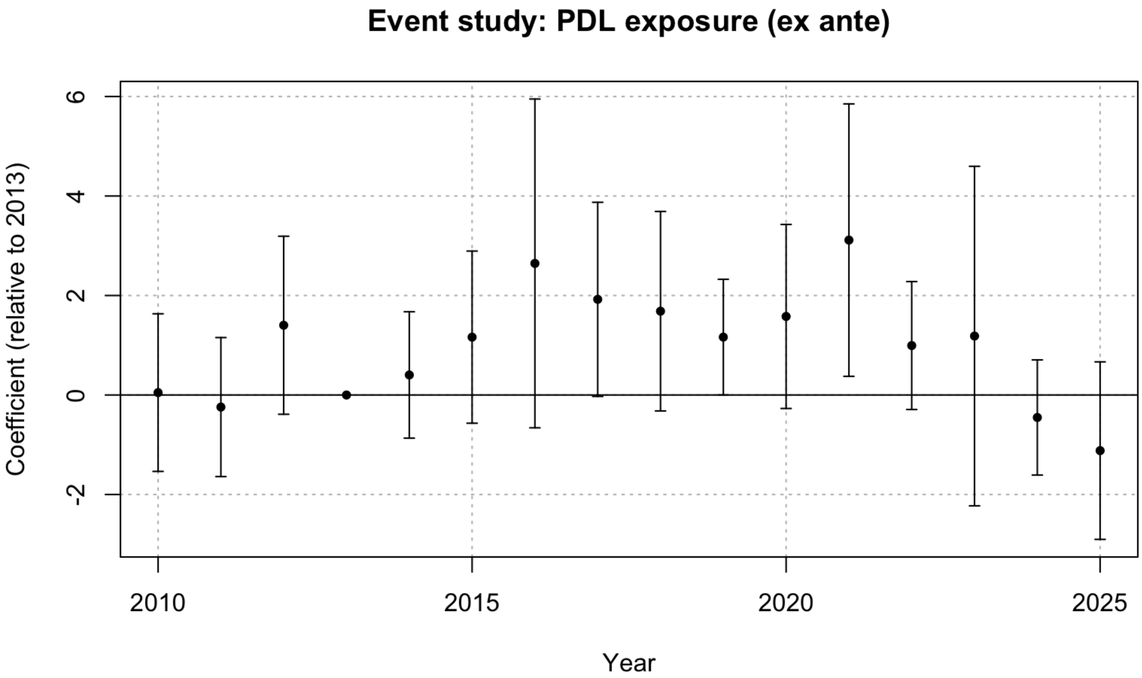

Figure 6: FEOLS event study: FDA approvals, PDL exposure (ex-ante)

The Medicaid exposure and PDL exposure coefficients from the joint specification are reported in Table 16. The results for Medicaid demand are consistent with the findings of the previous research question, with insignificant pre-period estimates and negative post-expansion coefficients growing from -0.086 in 2015 to -0.252 by 2024. Controlling for firm exposure to the PDL formulary yields larger Medicaid exposure coefficient estimates. This suggests that the RQ1 Medicaid coefficient partially absorbed variation associated with procurement exposure when PDL exposure was omitted. Including both measures therefore provides a cleaner comparison between the Medicaid demand response and the additional response associated with firms' exposure to PDL institutions. Since PDL exposure is a proxy for procurement exposure rather than a direct measure of rebate extraction, the estimates should not be interpreted as a full decomposition of the demand and pricing channels.

The PDL exposure coefficients point towards an additional source for the decline in post-expansion patenting. The pre-expansion PDL coefficients are positive but insignificant, providing support for the parallel trends assumption. The isolated positive estimate in 2014 is not persistent and should not be interpreted as a full effect of the ACA. From 2016 onwards, the PDL coefficients turn negative and become significant from 2019 onwards and grow to -6.925 by 2023. This indicates that firms with greater formulary exposure experienced an additional decline in patenting relative to the Medicaid demand effect.

This result should not be interpreted as showing that preferred placement itself reduces innovation incentives. The PDL exposure measure instead depicts the pre-expansion share of a firm's portfolio already on the PDL prior to Medicaid expansion, thus the negative coefficient indicates that firms more deeply embedded in Medicaid's formulary system before the reform experienced a stronger negative innovation response once Medicaid demand increased. This is consistent with these firms being more exposed to the procurement pressures that intensified following the expansion, even though securing preferred placement remains beneficial relative to failing one of the PDL stages.

Both the Medicaid exposure and PDL exposure coefficients for FDA approvals are statistically insignificant throughout the post-expansion period. There are isolated exceptions, with 2017 and 2018 for the Medicaid coefficients and 2019 and 2021 for the PDL coefficients, but these do not form a sustained pattern. Although a significant PDL effect is recorded for patents, the absence of a corresponding effect on FDA approvals is consistent with the broader interpretation that adjustment first appears in upstream R&D activity and has not yet propagated to the final stage of the R&D pipeline.

6.3 RQ3: Therapeutic-Class Heterogeneity

	Patents	FDA Approvals
Medicaid Share × ATC = A	−0.165***	1.211**
	(0.033)	(0.391)
Medicaid Share × ATC = B	0.134***	−0.396
	(0.024)	(2.567)
Medicaid Share × ATC = C	−0.123***	0.229
	(0.019)	(0.171)
Medicaid Share × ATC = D	−0.030	0.707**
	(0.108)	(0.270)
Medicaid Share × ATC = G	0.037	−2.516
	(0.056)	(2.666)
Medicaid Share × ATC = H	0.876***	1.144
	(0.136)	(1.358)
Medicaid Share × ATC = J	0.055	−0.145
	(0.054)	(0.165)
Medicaid Share × ATC = L	0.346	−0.321
	(0.452)	(2.789)
Medicaid Share × ATC = M	−0.336***	0.022
	(0.076)	(1.127)
Medicaid Share × ATC = N	−0.125***	0.034
	(0.028)	(0.074)
Medicaid Share × ATC = R	0.005	0.298
	(0.003)	(0.217)
Medicaid Share × ATC = S	0.093***	8.052*
	(0.017)	(3.737)
Num. obs.	19 617 728	561 600
R2	0.247	0.235

* p <0.05, ** p <0.01, *** p <0.001

Table 9: Triple-difference estimates: therapeutic-class heterogeneity (FEOLS, ex-ante)

The triple-difference estimates in Table 9 reveal substantial heterogeneity across therapeutic areas. Significant negative responses are observed in patenting behaviour for alimentary tract and metabolism, cardiovascular, musculo-skeletal, and nervous system drugs. RQ1 identifies an aggregate decline in patenting following the Medicaid expansion and these classes appear to account for a significant share of the overall response. Several of these classes rank highly in the annual NDC count averages reported in Table 17. Additionally, these classes also cover widely prescribed drug categories, such as diabetes and gastrointestinal medications, cardiovascular therapies, and antidepressants. This pattern supports the procurement mechanism developed in the theoretical model, where therapeutic areas with greater drug substitutability are exposed to more intense formulary competition and rebate pressure. Consequently, the procurement channel may more strongly offset the standard market-size incentive in these classes.

Positive patenting responses are observed for blood and blood-forming organs, hormonal preparations, and sensory organs. These classes are situated towards the lower end of the NDC count distribution, where therapeutic substitutability is weaker and the PDL channel is therefore less likely to dominate. The positive estimates are consistent with the standard market-size prediction in this setting, suggesting that the demand effect outweighs the disincentives associated with Medicaid's pricing pressure. Insignificant coefficients are observed for the remaining therapeutic classes, suggesting the demand and pricing pressure offset one another, or the expansion is unimportant for patenting decisions in these areas.

The presence of positive class-level estimates indicates that some therapeutic areas experience an innovation response that is driven largely by demand, suggesting that the standard market-size effect is not absent in the Medicaid setting. Instead, the aggregate negative patenting response identified in RQ1 appears to be concentrated in a subset of classes for which the procurement channel is likely to dominate. Hence, the heterogeneity analysis complements the key findings from RQ1 and the PDL-based evidence from RQ2, as opposed to implying that patenting responded uniformly across therapeutic classes in response to the ACA.

The FDA approval results contrast starkly with the patent pattern. Significant positive responses are observed for alimentary tract and metabolism, dermatologicals, and sensory organs, while no therapeutic class exhibits a significant negative response. The coefficient for sensory organs is large in magnitude but imprecisely estimated given the small number of approvals in that class.

Despite a strong negative patenting response in alimentary tract and metabolism drugs, positive FDA approval coefficients are observed in these areas in response to the ACA. This divergence is consistent with the pipeline interpretation, where FDA approvals observed during the post-expansion period most likely reflect R&D investments made prior to the Medicaid expansion, so the upstream patent decline has not yet propagated to the approval stage. Additionally, dermatologicals have a positive approval response while having an insignificant patent coefficient, indicating that approval estimates do not cleanly map onto patent responses within therapeutic areas. Both patents and FDA approvals have positive coefficients for pharmaceuticals relating to sensory organs, but the large standard errors suggest that this pattern should be interpreted with caution.

Several therapeutic classes display negative point estimates, but none reach statistical significance. This reinforces the earlier finding that the upstream decline in innovation has not yet reached the end stage of approvals. Consequently, the therapeutic-level heterogeneity identified in patenting has not yet been reflected in regulatory approvals, consistent with the long development horizons separating early-stage innovation from drug approval.

6.4 PDL Counterfactual Analysis

The estimates from the PDL counterfactual specification introduced in Section 4 are presented in Table 10, where the omitted baseline corresponds to a firm-class pair whose drugs do not pass either PDL stage.

	Patents	FDA Approvals
Post × P&T Pass	−11.399*	1.002
	(4.663)	(0.633)
Post × Price Pass	−7.386***	0.057
	(2.173)	(0.881)
Post × In PDL	15.549**	−0.112
	(5.215)	(1.174)
Num. obs.	5 281 696	151 200
R2	0.535	0.609

* p <0.05, ** p <0.01, *** p <0.001

Table 10: PDL counterfactual estimates

The patent results indicate that the innovation response to the ACA differs depending on a firm's pre-existing position within the PDL process. Firms that pass only one of the clinical or pricing stages experience reductions in patenting activity. Notably, the decline is larger for firms only passing the clinical stage, with a coefficient of −11.399, compared with a coefficient of −7.386 for firms that only pass the pricing stage. This indicates that exposure to either stage in isolation is associated with a post-expansion decline in patenting. Specifically, firms that only pass the clinical stage develop pharmaceutical products that meet the standards for safety and efficacy but do not secure preferred placement. However, firms which only pass the pricing stage are price-competitive but do not meet clinical criteria. In both cases, the expanded Medicaid market raises the value of PDL placement without granting these firms access to the associated demand.

The clinical stage measure being more negative indicates that the strongest post-Medicaid-expansion decline in patenting is associated with firms whose drugs pass the clinical threshold but are not cost competitive enough to secure preferred placement. Since these firms are unable to meet the pricing condition needed for preferred status on the Medicaid formulary, they cannot leverage their clinical eligibility for PDL access and are therefore the most exposed to the procurement channel. However, firms only passing the pricing stage produce cost-competitive drugs which do not meet the clinical criteria for preferred placement, and were consequently not positioned to secure PDL access regardless of pricing. Thus, PDL selection is less binding for this group, matching the smaller magnitude of the corresponding coefficient.

The coefficient for firms passing both stages of the PDL is large and positive, indicating that securing preferred placement substantially offsets the negative responses associated with passing either stage in isolation. Consequently, the total post-expansion effect for a firm whose portfolio passes both stages is −3.236. The net effect therefore remains negative and indicates a modest decline in patenting and is substantially smaller than the decline observed for firms only passing the clinical or pricing stages. Therefore, securing preferred placement appears to mitigate the negative innovation response, even if it does not fully reverse it.

The FDA approval results from the counterfactual specification do not mirror patenting behaviour. Both the clinical-stage and pricing-stage coefficients are small and positive, while the joint coefficient is mildly negative at −0.112. None of the coefficients being statistically significant is consistent with the broader pattern identified in RQ1 and RQ3, where the decline in patents has not yet reached the downstream approval stage given the long development horizons involved in pharmaceutical innovation. Consequently, a firm's position within the PDL process is not yet associated with an observable shift in drug approvals.

7 Robustness and Sensitivity Analysis

The baseline results are estimated using FEOLS with an ex-ante Medicaid market share measure as a proxy for firm-class exposure to the Medicaid reform. This section examines the sensitivity of these findings to alternative specifications, estimation methods, and treatment of the continuous exposure variable. Each research question is addressed in turn.

7.1 RQ1: Overall Innovation Response

7.1.1 Ex-Post Medicaid Exposure

The baseline specification uses pre-expansion Medicaid market shares to measure treatment intensity. Ex-post market shares are constructed from realised Medicaid prescription shares in 2014-2015 as an alternative measure of exposure to the ACA. The window for ex-post shares is restricted to the first two years of the reform in order to capture initial shifts in demand without extending into periods where endogenous response in firm entry or prescribing behaviour may contaminate the exposure measure. Estimates using ex-post exposure act purely as a robustness check since the measure would still reflect the actual post-expansion demand environment as opposed to being a predetermined exposure measure.

Under the estimated ex-post measure (Table 18), the patent coefficients are nearly identical to the ex-ante estimates. The pre-expansion coefficients remain small and insignificant, highlighting robustness to the parallel trends assumption while the post-reform coefficients follow the same gradual negative trajectory, growing from -0.088 in 2015 to -0.250 by 2024 before dropping slightly to -0.212 in 2025. These estimates are consistent with the ex-ante result, which also has a 2024 peak at -0.252. Similarly, the coefficients for FDA approvals are also in line with the baseline specification, where the coefficients are largely statistically insignificant with the exception of 2017 and 2019, while having no clear directional pattern. The alignment between the ex-ante and ex-post results therefore indicates that the findings are orthogonal to the choice of exposure measure.

7.1.2 FE Poisson Estimation

The baseline FEOLS specification models patent counts and FDA approvals linearly, although both outcomes are non-negative integer variables with substantial counts of zero. Consequently, the specification is re-estimated using a fixed-effects Poisson model, which therefore models the conditional mean of each of the innovation outcomes non-linearly. This allows for a robustness check on whether the baseline findings are sensitive to the linear treatment of the count outcomes, since the Poisson estimator remains consistent even when the outcome is overdispersed provided the conditional mean is correctly specified (Wooldridge, 1999). The resulting ex-ante and ex-post estimates are reported in Tables 19 and 20 respectively.

The FE Poisson estimates for patents are consistent with the FEOLS results. Under the ex-ante specification, the pre-expansion coefficients are small and insignificant, providing support for the parallel trends assumption. The post-expansion coefficients are negative and grow in magnitude over time, peaking at -1.474 by 2024 and falling slightly in 2025, consistent with the FEOLS findings. Therefore, at its peak in 2024, a ten percentage point increase in Medicaid exposure corresponds to an approximate 14.7 percent decrease in patents on average. The ex-post FE Poisson estimates follow a similar trajectory, with the 2024 peak being more negative at -2.021.

While the FDA approval estimates under the Poisson model are uniformly negative, they still remain insignificant across both the ex-ante and ex-post specifications. This is a slight deviation from the baseline OLS results, where the FDA estimates point in opposite directions, but both specifications still agree that the innovation response at the approval stage remains statistically insignificant. The findings for both innovation outcomes are consistent between the linear and Poisson specifications, suggesting that the post-expansion patterns are not sensitive to modelling the count outcomes linearly.

7.1.3 Binned Treatment Intensity

The baseline specification treats exposure to the ACA as a continuous variable. As an alternative test, therapeutic classes are grouped into low, medium and high exposure bins, allowing for separate event-study coefficients to be estimated across different parts of the Medicaid exposure distribution. This relaxes the assumption of linearity in treatment effects imposed by the continuous specification, and examines whether the post-expansion response is concentrated among classes which are more exposed to the Medicaid reform.

Therapeutic classes are grouped into the three bins based on their ex-ante Medicaid market shares, as reported in Table 21. Nervous system and cardiovascular drugs form the high-exposure bin, as their Medicaid market shares are substantially larger than other class shares. An approximate threshold of 10% is used to distinguish between the medium and low bin classes, thus anti-infectives and dermatologicals form the medium-exposure bin.

The low exposure bin is omitted from the regression, so the coefficients for the high and medium bins listed in Table 22 are estimated relative to the low-exposure bin. The pre-expansion coefficients for both the high and medium-exposure patent bins are small and statistically insignificant, supporting the parallel trends assumption within each group. While the medium-exposure bin largely contains positive insignificant coefficients throughout the post-expansion period with a peak decline of -0.02 in 2025, the high-exposure bin exhibits a growing negative trajectory with coefficients reaching -0.155 by the same year. This reinforces the finding from RQ3 that the post-expansion patenting decline is driven by classes at the upper end of the Medicaid exposure distribution. Importantly, the relative magnitude of the high and medium exposure coefficients is disproportionately larger than their respective Medicaid market shares. On average, exposure of the high bin is approximately 2.5 times larger than the medium bin, yet the peak patent responses approximately differ by a factor of eight. This suggests that the relationship between Medicaid exposure and patenting may be convex, with the response intensifying at greater levels of exposure as opposed to scaling proportionally.

FDA approval estimates are statistically insignificant for both the high and medium-exposure bins throughout the sample period, and the pre-expansion coefficients are close to zero in both bins, supporting the parallel trends assumption. The absence of significant post-expansion effects for FDA approvals is consistent with the baseline findings and reinforces the interpretation that the innovation response has not yet propagated to the downstream approval stage. Unlike the patent results, the estimates from the FDA binned specification provide no evidence of non-linear treatment effects across the Medicaid market share distribution, but this is driven by the absence of a detectable effect within each of the exposure bins rather than providing support for a linear relationship.

7.1.4 Continuous Difference-in-Differences

The binned event-study specification allowed for treatment effects to vary across the discrete groups, but the results were contingent on the boundaries specified for each exposure bin. Given this, the continuous difference-in-differences (Continuous DiD) estimator proposed by Callaway et al. (2024) is applied using the `contdid` R package (Callaway et al., 2025). This estimator non-parametrically identifies dose-specific treatment effects at each exposure-level under the parallel trends assumption, thereby avoiding the need to discretise the treatment variable. Conventional TWFE estimators with continuous treatment can be difficult to interpret when treatment effects differ across the distribution of exposure level. The Continuous DiD estimator avoids this by aggregating the dose-specific estimates into event-study coefficients and assesses pre-trends and post-treatment dynamics. The resulting coefficient estimates are reported in Table 23.

The Continuous DiD estimates for patents are consistent with the baseline results. The pre-expansion coefficients are small and statistically insignificant, supporting the parallel trends assumption. Notably, the coefficients are persistently significant from 2019 onwards, which is four years later than the 2015 onset under the baseline FEOLS model. This implies that the innovation response was not immediate as suggested by the baseline results, instead firms required several years to fully adjust their R&D decisions in response to the expansion.

The FDA approval estimates under the Continuous DiD model are statistically insignificant throughout the entire period. The pre-period estimates are small and fluctuate around zero, while the post-expansion estimates are mostly positive. The absence of a clear FDA approval response to the Medicaid reform is therefore robust to modelling Medicaid exposure non-parametrically within the Continuous DiD framework.

Taken together, the results from the sensitivity analysis show that the main findings are robust to discretising market shares into exposure bins and using a non-parametric continuous DiD approach. Crucially, the sustained negative patent response remains visible under an estimator developed specifically for continuous-treatment DiD designs, while drug approvals do not appear to have a clear response to the ACA. The Continuous DiD estimates are also consistent with the binned analysis for highly exposed patents, while bins with medium exposure do not display a statistically significant response. This pattern is consistent with the class-level heterogeneity identified in RQ3, where nervous system and cardiovascular drugs were among the classes experiencing the greatest patent reductions.

7.2 RQ2: Medicaid Demand and PDL Exposure

7.2.1 Ex-Post Medicaid Exposure

The second specification jointly estimates the innovation response to Medicaid demand exposure and exposure to the PDL. Like RQ1, an ex-post market share measure is used to determine whether the estimates are sensitive to the construction of the exposure variable, with the results reported in Table 24.

Once again, the ex-post Medicaid exposure coefficients for patents are nearly identical to the baseline results. The same reductions in Medicaid market share coefficients are observed, with the PDL exposure coefficient path also coinciding with the ex-ante model which saw negative coefficients from 2016 with significance from 2019. The pre-expansion coefficients for both the Medicaid and PDL measures remain small and statistically insignificant, supporting the parallel trends assumption even under the ex-post exposure construction.

The FDA approval coefficients under the ex-post specification are largely statistically insignificant for both the Medicaid and PDL exposure measures, with isolated incidences of significance in a small number of post-expansion years. This aligns with the baseline finding that the innovation response has not yet reached the final stage of the R&D pipeline. The agreement between the ex-ante and ex-post results therefore indicates that the decomposition into Medicaid demand and PDL exposure is not sensitive to the construction of exposure to Medicaid.

The FE Poisson, binned treatment, and continuous difference-in-differences estimators are not applied to the RQ2 specification. The joint estimation of two exposure variables under these alternative specifications is computationally demanding, and could not be completed with the device used for this study. Given this constraint, the alternative estimators are only used for the RQ1 specification where they are most informative for the aggregate response to the policy.

7.3 RQ3: Therapeutic-Class Heterogeneity

7.3.1 Ex-Post Medicaid Exposure

The RQ3 specification allows for heterogeneity in innovation behaviour to vary at ATC Level 1 therapeutic class. The ex-post exposure measure of Medicaid exposure is used as a robustness check.

The ex-post estimates (Table 25) are broadly consistent with the ex-ante baseline. Therapeutic classes that exhibited significant negative patent responses under ex-ante exposure, continue to display significant negative coefficients under the ex-post measure. Additionally, positive patent responses are still observed for therapeutic classes relating to blood related drugs, hormonal preparations, and sensory organs. Similarly, the FDA approval coefficients are aligned with their ex-ante results - significant positive responses for alimentary tract and dermatologicals, and no classes are recorded to have a significant reduction in approvals. The alignment between the two exposure measures indicates that the baseline class-level heterogeneity in innovation response is not driven by the choice of exposure measure.

7.3.2 Parallel Trends

The RQ3 triple-difference specification does not take the form of an event study, as the estimation of class-specific dynamic coefficients would be computationally intensive. Consequently, pre-trend coefficients cannot be estimated directly, so the parallel trends assumption cannot be tested in the same manner as RQ1 and RQ2. Since ATC class V is omitted from the regressions, it is the reference group for the DDD models. In this context, this specifically indicates that absent the ACA, the relationship between Medicaid exposure and changes in innovation outcomes would have evolved similarly across therapeutic classes relative to class V. The results from the earlier analyses and the model structure can provide evidence to support the plausibility of this assumption.

First, the RQ1 and RQ2 event studies consistently display small and statistically insignificant pre-expansion coefficients across all specifications, including FEOLS, FE Poisson, binned treatment and the continuous DiD approach. Therefore, therapeutic classes with differing levels of Medicaid exposure followed similar innovation trajectories prior to the ACA expansion. Since the same Medicaid exposure measure and firm-class panels are used for RQ3, the evidence for parallel trends from RQ1 and RQ2 provides support for the key identifying assumption of RQ3.

Second, as discussed in the methodology, a violation of parallel trends would require an additional shock around 2014 that differentially affected innovation within therapeutic classes such that it was correlated with exposure to Medicaid. There are no clear candidates for such a confounding shock.

8 Theoretical Model

This section outlines a game-theoretical model for pharmaceutical innovation and pricing to explain the empirical puzzle observed in RQ1. The model captures the stochastic nature of R&D, the specific procurement mechanism (PDL bidding) used by Medicaid under each state, and how current procurement outcomes dictate future innovation incentives.

In reality, several manufacturers may compete within a therapeutic class, and more than one product may appear on a state's PDL to account for clinical variations, such as allergies. However, the game uses a two-firm simplification for illustrative purposes. The same mechanism extends to settings with more firms. The model is set up for a representative therapeutic class $c \in \mathcal{C}$, where $\mathcal{C}$ denotes the set of ATC Level 1 classes. This class-specific segregation is consistent with Sutton's (1998) characterisation of pharmaceutical R&D as operating across distinct therapeutic areas with limited knowledge spillovers across groups. Each class is characterised by a Medicaid market size M_c, a rebate function $\tau_c(M_c)$, and a degree of drug substitutability n_c.

Figure 7 summarises the extensive form of the game, where innovation decisions in period 1 are mapped to the pricing outcomes, subsequently determining the state a firm enters for future R&D investment in period 2.

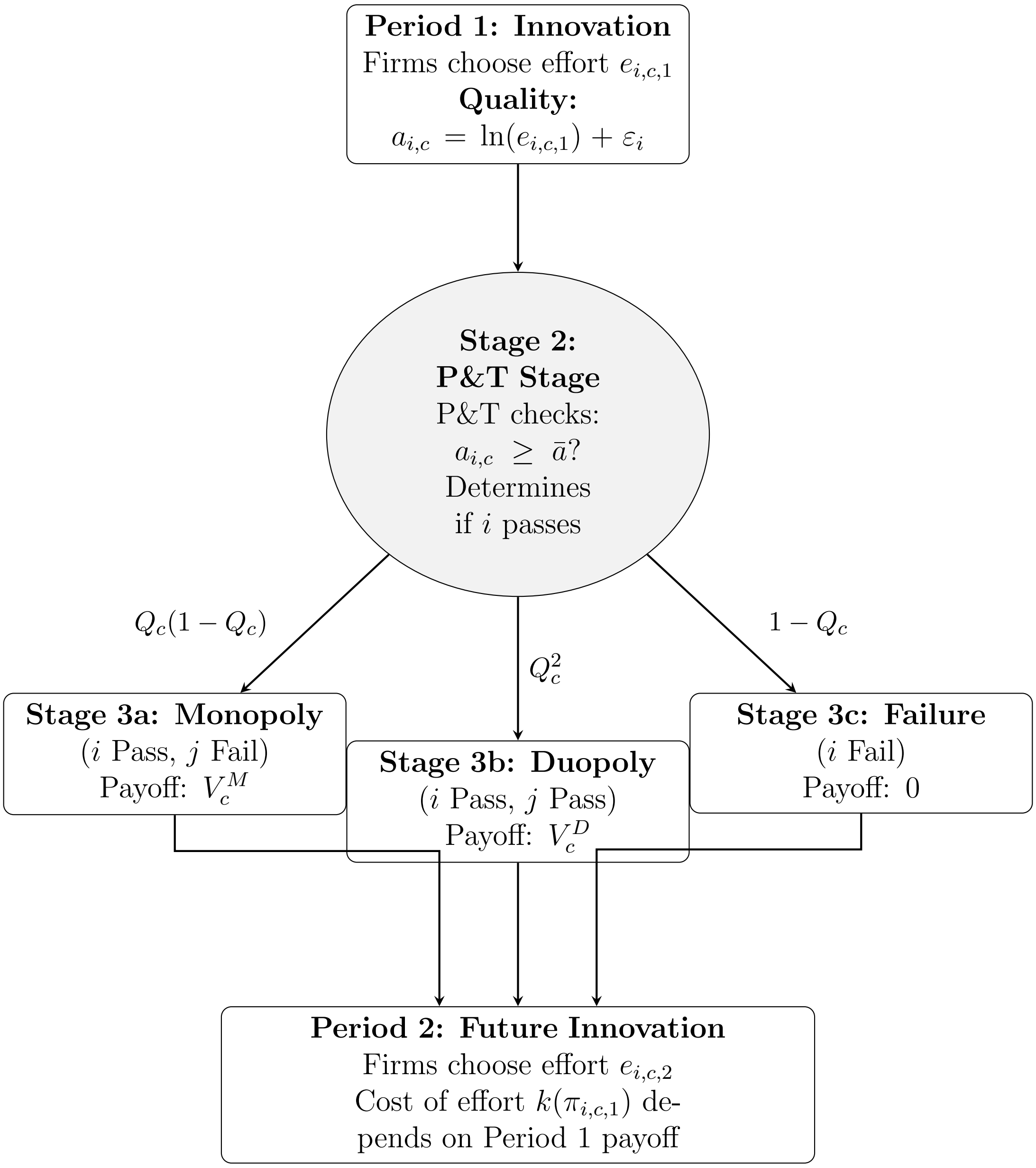

Figure 7: The Sequence of Play in Class c: Innovation, P&T Approval, Pricing, and Future Investment

8.1 Stage 1: The Innovation Decision (Effort)

First, firms make investment decisions for a drug they wish to market within therapeutic class c. Innovation is modelled to require effort, which raises the probability of clinical success while being costly, consistent with Sutton characterising pharmaceutical R&D as a search process where greater research effort raises the likelihood of success.

8.1.1 The Production Technology

The realised clinical quality $a_{i,c}$ is a function of the effort exerted by the firm and a stochastic component ε_i:

$$a_{i,c} = \ln(e_{i,c}) + \varepsilon_i, \qquad \varepsilon_i \sim N(0, \sigma^2) \tag{10}$$

where ε_i represents the inherent uncertainty of the R&D process.

8.1.2 The Cost of Effort

Firms i and j simultaneously choose R&D effort levels $e_{i,c} \geq 0$ within class c. Effort is assumed to be costly and convex:

$$C(e_{i,c}) = \frac{1}{2} k\, e_{i,c}^2 \tag{11}$$

8.2 Stage 2: The P&T Stage

Once effort is exerted it is now a sunk cost, nature determines the quality of each firm's drug ($a_{i,c}$), and therefore if it passes the P&T stage.

8.2.1 The Approval Condition

A drug is clinically approved (passes the P&T stage) if its quality $a_{i,c}$ exceeds a regulatory threshold quality $\bar{a}$:

$$\text{P\&T Stage}_{i,c} = \begin{cases} \text{Approved} & \text{if } a_{i,c} \geq \bar{a} \\ \text{Not Approved} & \text{if } a_{i,c} < \bar{a} \end{cases} \tag{12}$$

8.2.2 The Probability of Success (Q_c)

Since the random idiosyncratic shock ϵ_i takes on some normal distribution, the probability of a drug passing the P&T stage for a given class c is given as:

$$\begin{aligned} Q_c(e_{i,c}) &= \Pr\left(\ln(e_{i,c}) + \varepsilon_i \geq \bar{a}\right) && (13) \\ &= \Pr\left(\varepsilon_i \geq \bar{a} - \ln(e_{i,c})\right) && (14) \\ &= 1 - \Pr\left(\varepsilon_i < \bar{a} - \ln(e_{i,c})\right) && (15) \\ &= 1 - \Phi\left(\frac{\bar{a} - \ln(e_{i,c})}{\sigma}\right) && (16) \end{aligned}$$

where $\Phi(\cdot)$ is the standard normal CDF.

This defines the transition probabilities into Stage 3.

8.3 Stage 3: The Pricing Subgame

Firms that pass Stage 2 compete for PDL preferred status within their respective therapeutic class c. Each class c faces three distinct characteristics:

(i) A Medicaid market size M_c, which increased by $\Delta M_c \geq 0$ following the Medicaid expansion. Classes with higher Medicaid market shares experience greater shocks.

(ii) A rebate function $\tau_c(M_c)$ with $\tau_c' > 0$. This determines how aggressive supplemental rebates from Medicaid are at a per-prescription level.

(iii) A measure of therapeutic substitutability $n_c \geq 1$. Formally, this is defined as the number of clinically interchangeable drugs available within class c.

The model does not impose a functional form on τ_c - the only requirement is that $\tau_c' > 0$. Additionally, the steepness of the rebate schedule is linked to the degree of therapeutic substitutability, that is:

$$\frac{\partial \tau_c(M_c, n_c)}{\partial n_c} > 0 \tag{17}$$

When the PDL committee has many clinically interchangeable drugs to choose from, it can credibly threaten to exclude any given drug unless a deeper rebate is offered. Therefore, classes with more substitutes face steeper rebate schedules.

Stage 3 of the game takes on one of three mutually exclusive states.

8.3.1 Stage 3a: The Monopoly State

Condition: Firm i passes the P&T stage while firm j fails:

$$a_{i,c} \geq \bar{a} > a_{j,c}$$

Mechanism: Firm i is the sole player in class c and negotiates directly with Medicaid.

Setup The firm faces a class-specific market demand M_c which is perfectly inelastic with respect to price. This assumption is made since Medicaid enrollees are fully insured. However, the state imposes a maximum allowable price $\bar{p}$ and the firm also pays the class-specific rebate $\tau_c(M_c)$.

Optimisation Problem The firm chooses price p to maximise profits:

$$\max_p \pi_{i,c} = M_c(p - c_i - \tau_c(M_c)) \quad \text{s.t.} \quad p \leq \bar{p} \tag{18}$$

Since demand is inelastic, the profit function is strictly increasing in p.

Equilibrium Result The firm sets the price at the binding constraint: $p^* = \bar{p}$, thus the resulting profit is:

$$\pi_{i,c}^{Mono}(c_i) = M_c(\bar{p} - c_i - \tau_c(M_c)) \tag{19}$$

Since marginal costs c_i are unknown until stage 3 is reached, the expected profits are calculated across marginal cost states:

$$V_c^M = \mathbb{E}_c[\pi_{i,c}^{Mono}] \tag{20}$$

8.3.2 Stage 3b: The Duopoly State (Auction)

Condition: Both firms pass the P&T stage in class c:

$$a_{i,c} \geq \bar{a}, \qquad a_{j,c} \geq \bar{a}$$

Mechanism: Firms compete in a first-price sealed-bid procurement auction.

Setup Each firm independently draws a marginal cost c_i, c_j from a common distribution $F(c)$ with support $[\underline{c}, \bar{c}]$. Each firm then submits a sealed bid b_i, b_j which is equivalent to the price that the drug is sold to Medicaid at. The lowest bidder wins PDL status and, for that class, obtains the full Medicaid market volume M_c minus the class-specific statutory wedge $\tau_c(M_c)$.

$$\pi_{i,c} = \begin{cases} M_c\big(b_i - c_i - \tau_c(M_c)\big) & \text{if } b_i \leq b_j, \\ 0 & \text{if } b_i > b_j. \end{cases}$$

Equilibrium Derivation Assume that there exists a strictly increasing and differentiable symmetric Bayesian-Nash Equilibrium bidding strategy $b_c(c)$.

The Probability of Winning Suppose firm i has cost c but considers submitting a bid $\tilde{b}$ corresponding to the optimal bid of type t (i.e., $\tilde{b} = b_c(t)$). Firm i wins if and only if its bid is lower than the rival's bid $b_c(c_j)$:

$$b_c(t) < b_c(c_j) \iff t < c_j \tag{21}$$

Therefore, the probability of winning the bidding game is given as:

$$\Pr(\text{win}) = \Pr(c_j > t) = 1 - F(t) \tag{22}$$

The Optimisation Problem The firm chooses the optimal type t to maximise expected profit:

$$\max_t \mathbb{E}[\pi_{i,c}(t,c)] = M_c(b_c(t) - c - \tau_c(M_c))[1 - F(t)] \tag{23}$$

Let $U_c(c)$ be the value function (maximum expected profit) for a firm with true cost c within class c:

$$U_c(c) = \max_t M_c(b_c(t) - c - \tau_c(M_c))[1 - F(t)] \tag{24}$$

Applying the Envelope Theorem Differentiating the value function $U_c(c)$ with respect to cost c yields the derivative of the objective function evaluated at the optimal choice $t = c$, corresponding to truthful reporting:

$$U_c'(c) = \left.\frac{\partial \mathbb{E}[\pi_{i,c}(t,c)]}{\partial c}\right|_{t=c} = -M_c[1 - F(c)] \tag{25}$$

Solving for Expected Profit Using the fact that the highest-cost type earns zero profit, $U_c(\bar{c}) = 0$, integrating $U_c'(c)$ from c to $\bar{c}$ gives:

$$U_c(c) = M_c \int_c^{\bar{c}} (1 - F(u))\, du. \tag{26}$$

Recovering the Bidding Function Equating (26) with the firm's expected profit at the optimum gives the symmetric equilibrium bidding strategy:

$$b_c(c) = c + \tau_c(M_c) + \frac{\int_c^{\bar{c}} (1 - F(u))\, du}{1 - F(c)}. \tag{27}$$

Equilibrium Result Since $U_c(c)$ is the firm's equilibrium expected payoff, the expected profit for a firm of type c in the duopoly state is:

$$\Pi_c^{Duo}(c) = U_c(c) = M_c \int_c^{\bar{c}} (1 - F(u))\, du. \tag{28}$$

The expectation of the duopoly state is calculated over marginal costs, as V_D:

$$V_c^D = \mathbb{E}_c[\Pi_c^{Duo}]. \tag{29}$$

This implies that V_c^D increases linearly with M_c but is not directly affected by the rebate wedge $\tau_c(M_c)$. This occurs since the rebate enters as a common cost component in the auction and is therefore reflected in equilibrium bids. Consequently, the rebate wedge reduces V_c^M directly, but does not reduce V_c^D in the same way.

8.3.3 Stage 3c: The Failure State

Condition: Firm i fails in class c:

$$a_{i,c} < \bar{a}$$

The firm fails to meet the clinical threshold required for PDL consideration, therefore there is no opportunity to sell to the Medicaid market.

Equilibrium Result The firm does not enter the market and earns zero revenue.

$$\pi_{i,c}^{Fail} = 0 \tag{30}$$

8.4 Equilibrium Analysis

Solve for equilibrium effort e^* within each therapeutic class c, and subsequently model how current procurement outcomes shape future innovation.

8.4.1 The Maximisation Problem

Within class c, the firm chooses effort $e_{i,c}$ to maximise total expected profit, where $\Psi_{i,c}$ is the probability-weighted sum of payoffs in each state minus the cost of effort:

$$\max_{e_{i,c}} \Psi_{i,c} = \underbrace{Q_c(e_{i,c})[1 - Q_c(e_{j,c})]V_c^M}_{\text{Stage 3a}} + \underbrace{Q_c(e_{i,c})Q_c(e_{j,c})V_c^D}_{\text{Stage 3b}} - \frac{1}{2} k\, e_{i,c}^2 \tag{31}$$

where V_c^M and V_c^D are the class-specific expected profits derived in the monopoly and duopoly cases, respectively.

8.4.2 Optimal Effort (e_c^*)

The First Order Condition (FOC) with respect to effort $e_{i,c}$ is derived by differentiating $\Psi_{i,c}$:

$$\frac{\partial \Psi_{i,c}}{\partial e_{i,c}} = Q_c'(e_{i,c})(1 - Q_c(e_{j,c}))V_c^M + Q_c'(e_{i,c})Q_c(e_{j,c})V_c^D - k\, e_{i,c} = 0 \tag{32}$$

This yields the following expression, where the marginal benefit of effort is equal to the associated marginal cost:

$$Q_c'(e_{i,c}) \left[(1 - Q_c(e_{j,c}))V_c^M + Q_c(e_{j,c})V_c^D\right] = k\, e_{i,c} \tag{33}$$

The term in square brackets is the expected payoff contingent on P&T approval within class c. In a symmetric equilibrium where $e_{i,c} = e_{j,c} = e_c^*$, this implicitly defines the class-specific optimal effort level e_c^*.

For this to be a strict global maximum, the Second Order Condition (SOC) must be strictly negative. Differentiating the FOC again with respect to $e_{i,c}$ yields:

$$\frac{\partial^2 \Psi_{i,c}}{\partial e_{i,c}^2} = Q_c''(e_{i,c}) \left[(1 - Q_c(e_{j,c}))V_c^M + Q_c(e_{j,c})V_c^D\right] - k < 0 \tag{34}$$

Since the term in square brackets is positive and $k > 0$, concavity in Q_c is sufficient for a global maximum, i.e. $Q_c''(e_{i,c}) < 0$.

8.4.3 Strategic Substitutes

The strategic relationship between the firms' efforts within class c can be determined by analysing the slope of the best response function.

Let $e_{i,c}^* = BR_{i,c}(e_{j,c})$ be the implicitly defined best response function that solves the FOC:

$$\frac{\partial \Psi_{i,c}(BR_{i,c}(e_{j,c}), e_{j,c})}{\partial e_{i,c}} = 0 \tag{35}$$

Totally differentiating this with respect to $e_{j,c}$ yields:

$$\frac{d}{de_{j,c}} \left[\frac{\partial \Psi_{i,c}(BR_{i,c}(e_{j,c}), e_{j,c})}{\partial e_{i,c}}\right] = 0 \tag{36}$$

$$\iff \quad \frac{\partial^2 \Psi_{i,c}}{\partial e_{i,c}^2}\frac{dBR_{i,c}(e_{j,c})}{de_{j,c}} + \frac{\partial^2 \Psi_{i,c}}{\partial e_{i,c}\partial e_{j,c}} = 0 \tag{37}$$

$$\iff \quad \frac{dBR_{i,c}(e_{j,c})}{de_{j,c}} = -\frac{\frac{\partial^2 \Psi_{i,c}}{\partial e_{i,c}\partial e_{j,c}}}{\frac{\partial^2 \Psi_{i,c}}{\partial e_{i,c}^2}} \tag{38}$$

Since the denomininator is the SOC (which is known to be strictly negative), the sign of the best response derivative depends purely on the sign of the cross-partial derivative:

$$\frac{\partial^2 \Psi_{i,c}}{\partial e_{i,c}\partial e_{j,c}} = Q_c'(e_{i,c}) \left[-Q_c'(e_{j,c})V_c^M + Q_c'(e_{j,c})V_c^D\right] = Q_c'(e_{i,c})Q_c'(e_{j,c})[V_c^D - V_c^M] \tag{39}$$

Additionally, the monopoly value strictly exceeds the duopoly value ($V_c^M > V_c^D$), and marginal probabilities are positive ($Q_c' > 0$), thus the cross-partial derivative is strictly negative.

Given both the SOC and the cross-partial derivatives are negative, it follows that the slope of the best response function is negative ($\frac{dBR_{i,c}}{de_{j,c}} < 0$). Consequently, within each class c, the R&D efforts of the two firms can be classed as strategic substitutes.

If firm j raises its effort in class c, it is likelier to pass clinical trials and the corresponding P&T stage. Therefore, for any given effort level for $e_{i,c}$, the probability of both firms entering the duopoly auction stage rises, and in such a case minimal profits (V_c^D) are obtained. Since the expected profits shrink under such a situation, the marginal benefit of firm i's effort decreases, thereby incentivising it to best-respond by lowering its own effort.

8.4.4 The "Incentive Trap" Mechanism

This section now aims to derive the effect of the expansion of Medicaid ($M_c \uparrow$) on innovation effort e_c^* within each class. A necessary condition for a reduction in innovation within class c is if the marginal benefit of effort decreases with class-specific market size.

The Mechanism

1. **Stage 3a (Monopoly):** From Equation (19), the value of the monopoly state in class c is $V_c^M = M_c(\bar{p} - \bar{c} - \tau_c(M_c))$. The effect of expansion on the monopoly value is:
$$\frac{dV_c^M}{dM_c} = \underbrace{(\bar{p} - \bar{c} - \tau_c)}_{\text{Volume Effect (+)}} - \underbrace{M_c \, \tau_c'(M_c)}_{\text{Monopsony Effect (-)}} \tag{40}$$

2. **Stage 3b (Duopoly):** From Equation (28), V_c^D increases with M_c, thus the expansion unambiguously raises the value of the duopoly setting.

Define M_c^* as the threshold by which the value of the monopoly state decreases with market size. This is defined implicitly by the condition:

$$(\bar{p} - \bar{c} - \tau_c(M_c^*)) - M_c^* \, \tau_c'(M_c^*) = 0 \tag{41}$$

or equivalently
$$(\bar{p} - \bar{c} - \tau_c(M_c^*)) = M_c^* \, \tau_c'(M_c^*) \tag{42}$$

At M_c^*, the marginal benefit of selling one additional drug to Medicaid is exactly equal to the marginal rebate extraction imposed by Medicaid. A unique interior solution exists whenever margins are initially positive ($\bar{p} - \bar{c} > \tau_c(0)$) and the rebate extraction dominates as the Medicaid market grows ($\lim_{M_c \to \infty} \tau_c'(M_c) = \infty$).

The incentive trap binds within class c if and only if the post-expansion market size exceeds this threshold, that is:

$$M_c^{post} \equiv M_c^{pre} + \Delta M_c > M_c^* \tag{43}$$

Since the rebate function τ_c reflects the competitive structure of a therapeutic drug class, the threshold market-size M_c is class-specific.

Result 1 (Substitutability lowers the threshold M_c^*)**.** *Based on Equation 17 $\tau_c'(M_c)$ is increasing in n_c, so classes with more substitutes have a lower threshold M_c^*. Therefore, classes with more interchangeable drugs are likelier to meet the threshold market size and become trap-bound.*

Proof. The threshold M_c^* is defined by:

$$(\bar{p} - \bar{c} - \tau_c(M_c^*)) = M_c^* \tau_c'(M_c^*)$$

An increase in the number of substitutes n_c makes the rebate schedule steeper and therefore $\tau_c'(M_c)$ increases. This raises the marginal cost of the rebate $M_c\tau_c'(M_c)$, while also reducing remaining revenues as the stronger bargaining position allows for Medicaid to extract a larger rebate $\tau_c(M_c)$.

Therefore, at any given market size M_c, higher substitutability makes rebate extraction larger and revenues smaller, making the threshold level reached for a lower market size M_c. Thus:

$$\frac{dM_c^*}{dn_c} < 0$$

□

Result 2 (Larger shocks increase the probability of crossing the threshold)**.** *Classes with higher pre-expansion Medicaid market share experience larger increases in market size ΔM_c following the ACA expansion. This makes it likelier for these markets to cross the threshold: $M_c^{post} > M_c^*$.*

Therefore, Results 1 and 2 imply that the decline in innovation should be concentrated in classes with high Medicaid exposure and high levels of substitutability. Specifically, high levels of Medicaid exposure raise M_c^{post}, while high substitutability lowers M_c^* by making the rebate schedule steeper.

Classification of therapeutic classes The FOC (33) implies that equilibrium effort depends on the expected payoff contingent on passing the P&T stage

$$(1 - Q_c)V_c^M + Q_c V_c^D$$

Therefore, the sign of effort in response to the ACA Medicaid expansion depends on:

$$\frac{de_c^*}{d\Delta M_c} = (1 - Q_c)\frac{dV_c^M}{dM_c} + Q_c \frac{dV_c^D}{dM_c} \tag{44}$$

The duopoly value will always increase optimal efforts ceteris paribus. However, when the monopoly value is much larger than the duopoly value ($V_c^M \gg V_c^D$), the direction of optimal effort response is primarily driven by the monopoly channel. In this case, the relevant threshold market size can be approximated by the point at which the monopoly value (V_c^M) starts to fall:

$$\frac{de_c^*}{d\Delta M_c} \gtreqless 0 \quad \Longleftrightarrow \quad \frac{dV_c^M}{dM_c} \gtreqless 0 \quad \Longleftrightarrow \quad M_c^{post} \lesseqgtr M_c^* \tag{45}$$

Technically, M_c^* is the threshold market-size under the monopoly channel as opposed to the exact threshold for equilibrium effort. The effort threshold would be weakly greater than M_c^* since the duopoly channel still increases with M_c. Therefore, the derived threshold market size below should be interpreted as exact when the monopoly channel dominates.

This separates therapeutic classes into three separate categories:

- **Volume-dominated classes** ($M_c^{post} < M_c^*$): The expansion increases V_c^M so equilibrium effort rises - the standard market-size prediction holds.
- **Trap-bound classes** ($M_c^{post} > M_c^*$): The expansion decreases V_c^M. In the case of the monopoly channel dominating the duopoly channel, equilibrium effort falls.
- **Near-threshold classes** ($M_c^{post} \approx M_c^*$): The volume and monopsony effects offset each-other so the net effect of the expansion on innovation is insignificant.

The rent gap The difference between monopoly and duopoly payoffs matters for innovation incentives:

$$\Delta V_c \equiv V_c^M - V_c^D. \tag{46}$$

The rent gap ΔV_c is smaller in classes where the incentive trap binds. This is because higher drug substitutability gives Medicaid a stronger position for bargaining, thereby yielding a steeper rebate schedule which can reduce V_c^M. Consequently, two classes with the same post-expansion market size M_c can have different innovation responses - a class with more substitutes will have a smaller rent gap and lower associated levels of effort.

8.4.5 Institutional and Empirical Justification

Two crucial assumptions of this model are that the Medicaid rebate wedge, $\tau_c(M_c)$, is (i) strictly increasing in market size and (ii) its steepness is increasing within-class drug substitutability. While federal statutory rebates are fixed percentages, state-Medicaid providers negotiate supplemental rebates with manufacturers in exchange for PDL placement. Additionally, some states join Multi-State Purchasing Pools (such as the National Medicaid Pooling Initiative and the Sovereign States Drug Consortium), allowing them to leverage the population under multiple states to increase their effective market size (M_c). These coalitions are therefore able to tie the size of their negotiated rebate to the volume of pooled lives.

This assumption is tested using the SDUD. States are mapped to their multi-state purchasing pools to reflect the true negotiating market size within each ATC. Per-prescription rebates are regressed against the natural log of market size with and without the inclusion of the number of distinct NDCs within a class n_c, class and year fixed effects are included:

$$\text{Rebate}_{c,t} = \alpha + \beta_1 \log(M_c) + \beta_2 \log(n_c) + \lambda_c + \delta_t + \varepsilon_{c,t} \tag{47}$$

Table 11 shows that the coefficients on both $\log(M_c)$ and $\log(n_c)$ are positive and significant, indicating that larger Medicaid markets and a greater number of distinct drugs within a class are both associated with higher rebates. These results support the assumptions that $\tau_c'(M_c) > 0$ and that the rebate schedule steepens with n_c.

Table 11: Prescription rebate regressions

	Rebate	Rebate
Log market size	5.513***	4.408***
	(0.035)	(0.081)
Log NDCs		4.646***
		(0.309)
Num. obs.	11 305 354	11 305 354
R2	0.046	0.046

* p <0.05, ** p <0.01, *** p <0.001

8.4.6 State-Dependent Innovation

Empirically, firms that gain PDL placement tend to innovate more in subsequent periods. In order to capture this dynamic, the current game can be embedded in a dynamic framework where procurement outcomes influence future innovation incentives.

In this setting, period 1 procurement outcomes in class c determine the firm's position in Period 2. A firm may fail to gain access, gain access under competitive rents, or gain access under higher monopoly rents. A natural explanation for this state dependence is the internal-finance channel. Hall (2002) shows that R&D is especially exposed to financing frictions because it is risky and intangible, making external finance costly. This concern is amplified in pharmaceuticals, where drug development involves substantial sunk costs, highly variable returns, and informational asymmetries that disadvantage firms with limited internal resources (Sutton, 1998). Securing preferred access on the PDL generates additional cash flows that relax financing constraints, thereby lowering the effective marginal cost of future R&D.

Additionally, firms that pass the P&T stage and negotiate rebates may gain organisational knowledge specific to navigating Medicaid procurement. This lowers the effective cost of future innovation without relying on pharmaceutical discovery itself accumulating across research trajectories. This modelling choice is supported by Sutton (1998), who documents limited persistence in firms' follow-on drug successes. This makes cash flow and procurement-related capabilities a more appropriate basis for state-dependent innovation than persistent dominance in drug discovery.

This is consistent with dynamic models of competition. For example, Budd et al. (1993) show that a firm's current competitive position can shape future investment incentives, while Aghion et al. (2001) show that firms in near-symmetric states may invest strongly to escape competition. In this model, firms which do not gain preferred status may still have incentives to innovate in the next period, but they do so with a cost disadvantage relative to firms that secured PDL rents.

Formalising the Multi-Period Game Now allow the period 2 cost of effort parameter k to depend on the realised payoff from period 1. Note that the potential period 1 payoffs for firm i in class c are:

$$\pi_{i,c,1} \in \{V_c^M, V_c^D, 0\} \tag{48}$$

where $V_c^M > V_c^D > 0$.

Assuming that internal cash flow and accumulated regulatory capabilities lower the effective marginal cost of future R&D, the Period 2 effort cost function is:

$$C_{c,2}(e_{i,c,2}; \pi_{i,c,1}) = \frac{1}{2} k(\pi_{i,c,1})\, e_{i,c,2}^2, \quad k'(\pi_{i,c,1}) < 0 \tag{49}$$

Higher period 1 profits strictly lower the cost parameter thus:

$$k(V_c^M) < k(V_c^D) < k(0) \tag{50}$$

In period 2, firm i now solves an updated version of the FOC (Equation 33), where the new cost parameter $k_{i,c} = k(\pi_{i,c,1})$ is taken as given:

$$Q_c'(e_{i,c,2}) \left[(1 - Q_c(e_{j,c,2}))V_c^M + Q_c(e_{j,c,2})V_c^D\right] - k_{i,c}\, e_{i,c,2} = 0 \tag{51}$$

Apply the implicit function theorem in order to observe how optimal effort responds to the cost parameter $k_{i,c}$. To do this, let $\Psi_{i,c,2}$ denote firm i's objective function in period 2 for therapeutic class c. Differentiating the first-order condition with respect to $k_{i,c}$ gives

$$\frac{\partial e_{i,c,2}^*}{\partial k_{i,c}} = -\frac{\frac{\partial^2 \Psi_{i,c,2}}{\partial e_{i,c,2} \partial k_{i,c}}}{\frac{\partial^2 \Psi_{i,c,2}}{\partial e_{i,c,2}^2}} = -\frac{-e_{i,c,2}^*}{\frac{\partial^2 \Psi_{i,c,2}}{\partial e_{i,c,2}^2}} \tag{52}$$

The SOC implies that $\frac{\partial^2 \Psi_{i,c,2}}{\partial e_{i,c,2}^2} < 0$, and since optimal effort is positive, it follows that:

$$\frac{\partial e_{i,c,2}^*}{\partial k_{i,c}} < 0 \tag{53}$$

That is, optimal effort is strictly decreasing in the cost parameter.

By construction, the cost parameter is strictly decreasing in period 1 profits, thus higher period 1 profits lower the cost of period 2 effort and increase optimal period 2 efforts:

$$e_{c,2}^*(V_c^M) > e_{c,2}^*(V_c^D) > e_{c,2}^*(0) \tag{54}$$

Implications for the Incentive Trap This two-period structure now strengthens the main result of the model. If the Medicaid expansion raises rebate extraction and lowers V_c^M, then it reduces both current and future innovation incentives in class c. The expansion affects outcomes at the time of the policy by lowering the payoff to successful innovation, while weaker internal cash flows raise the cost of future R&D efforts. Consequently, lower current rents can lead to lower innovation in later periods, with the severity of this effect varying across classes.

Result 3 (PDL winners gain less in trap-bound classes)**.** *The advantage of PDL winners is weaker in trap-bound classes than in volume-dominated classes. To see this, consider two classes c and c', where $M_c^{post} > M_c^*$ and $M_{c'}^{post} < M_{c'}^*$:*

$$e_{c,2}^*(V_c^M) - e_{c,2}^*(0) < e_{c',2}^*(V_{c'}^M) - e_{c',2}^*(0)$$

In 'trap-bound' classes, the rebate wedge lowers the value of V_c^M, so PDL winners receive a smaller payoff advantage over firms that fail to gain preferred status. While the cost parameter ranking $k(V_c^M) < k(V_c^D) < k(0)$ still holds, the gaps between these cost parameters narrow. Therefore, PDL winners still invest more than non-winners, but the difference in future effort is smaller. However, classes which are volume-dominated classes have a larger relative monopoly value V_c^M, so preferred firms retain a greater cost advantage and therefore invest more relative to their rivals.

8.5 Model Predictions and Empirical Mapping

8.5.1 Comparative Statics

To characterise the predictions of the model, equilibrium effort is solved for numerically from the FOCs. Since no closed-form solution exists for e_c^*, the comparative statics are computed by varying one parameter and holding all else fixed at baseline values reported in Table 26. Since the functional form of $\tau_c(M_c)$ is unknown, three functional forms are proposed for the simulations: a power specification $\tau_c = \alpha_c M_c^{\beta_c}$, a logarithmic specification $\tau_c = \alpha_c \ln M_c$, and a linear specification $\tau_c = \alpha_c M_c$. The resulting comparative statics are shown in Figure 8.

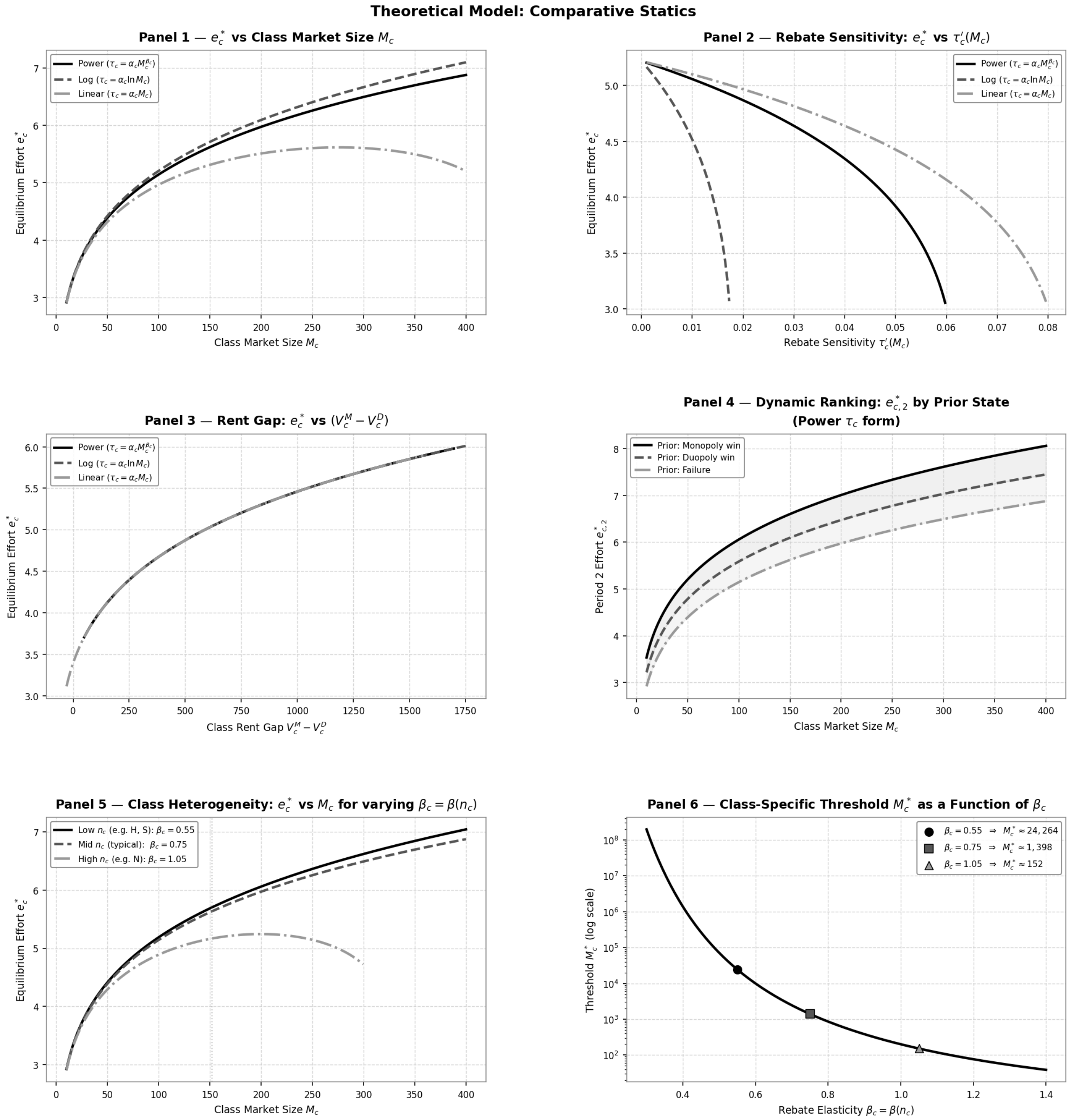

Figure 8: Comparative statics of equilibrium effort e_c^*. Baseline parameter values are reported in Table 26.

Panel 1: Equilibrium effort and market size. The first panel displays e_c^* as a function of market size M_c for each $\tau_c(M_c)$ specification. Under the logarithmic and power forms, effort rises monotonically with M_c indicating that the volume effect dominates the rebate effect, that is, the additional revenue from selling to a larger market dominates, thus firms invest more in R&D as the market expands. However, a hump-shaped profile is observed under the linear form. Effort initially increases with M_c but declines subsequently - this gives rise to the incentive trap mechanism. The linear rebate schedule implies that Medicaid extracts revenues one-for-one with market size. Once M_c is large enough, the term $M_c\tau_c'(M_c) = \alpha_c M_c$ dominates the volume gain $(\bar{p} - \bar{c} - \tau_c)$, so $dV_c^M/dM_c < 0$ and the marginal benefit of effort falls.

The empirical results from the triple difference model suggest that therapeutic classes differ in their position along the market-size axis and the steepness of their rebate schedules. Classes B, H, and S appear to be in the upward-sloping, volume-dominated region. Classes D, G, J, L, and R sit closer to the peak. Classes A, C, M, and N are high-volume and therapeutically competitive, and appear to sit past the peak in the trap-bound region.

Panel 2: Equilibrium effort and rebate sensitivity. The second panel keeps M_c fixed at $M_0 = 100$ and isolates the effect of Medicaid's monopsony power on equilibrium efforts by varying the rebate scale parameter α_c. Across all three rebate specifications equilibrium effort is strictly decreasing in $\tau_c'(M_c)$. However, the curves diverge substantially in their rate of decline. The logarithmic form appears to have the steepest curve, while the linear and power forms maintain higher efforts across a wider range of sensitivity values. Classes with more therapeutic substitutes (n_c) face steeper rebate sensitivity and therefore correspond to lower levels of effort e_c^*.

Panel 3: Equilibrium effort and the rent gap. The third panel varies the price cap $\bar{p}$ to isolate the effect of the rent gap ΔV_c on incentive efforts e_c^*. Innovation is observed to be driven by the gap between monopoly and duopoly payoffs, rather than by the level of either payoff alone. The three curves coincide since the FOC depends on this rent differential and the relationship between e_c^* and ΔV_c is independent of the specific rebate functional form. Importantly, this panel establishes that the reduction in ΔV_c is sufficient for the reduction in innovation and not the rise in market-size alone. Therefore, any policy that reduces value of the monopoly state relative to the duopoly state lowers equilibrium effort.

Panel 4: Period 2 effort by prior state. The fourth panel displays the rankings of period 2 efforts following period 1 outcomes under the power specification $\tau_c(M_c) = \alpha_c M_c^{\beta_c}$. Equilibrium efforts are computed under three cost parameters corresponding to the three possible period 1 outcomes: a monopoly win, a duopoly win, and failure. The following effort ranking hold for all market sizes:

$$e_{c,2}^*(V_c^M) > e_{c,2}^*(V_c^D) > e_{c,2}^*(0)$$

This is consistent with the idea that firms securing higher profits in period 1 face lower R&D costs in period 2 and therefore invest more in subsequent innovation. The effort gap between the three paths widen with market size, showing that the long-run innovation consequences of period 1 procurement outcomes become stronger in larger markets.

Panel 5: Drug substitutability and effort. The fifth panel displays how the relationship between market size and effort changes with drug substitutability. Three cases are shown: low substitutability (e.g. ATC classes H and S), intermediate substitutability, and high substitutability (ATC class N).

When drug substitutability is low, effort continues to rise with market size and the incentive trap does not bind. However, as more drug substitutes become available, the threshold M_c^* lowers, so the turning point for effort appears for lower Medicaid market sizes. This illustrates Result 1, where higher drug substitutability lowers M_c^* and makes the incentive trap more likely to bind.

Panel 6: The class-specific threshold. The sixth panel plots the threshold market-size M_c^* against the rebate elasticity β_c under the power specification. The threshold falls monotonically as elasticity increases. This indicates that classes with steeper rebate schedules require a much smaller Medicaid market size for equilibrium efforts to start lowering.

8.5.2 Theoretical States and Empirical Outcomes

To bridge the theoretical framework with the empirical analysis, the regression variables are mapped directly to the realised outcomes of the game.

Mapping to the PDL Counterfactuals Empirically, the pharmaceutical procurement pipeline is observed through the lens of three stages: passing clinical approval, submitting a pricing bid, and gaining preferred status on the PDL. These variables represent different ex-post paths for success:

- **Partial Success:** This represents firms that exert sufficient effort to clear the clinical stage ($a_{i,c} \geq \bar{a}$) but enter the duopoly bidding stage and lose due to presenting a higher price than the opposing firm. Clearing the clinical threshold provides these firms with some regulatory know-how, but failing to capture the market means that they do not receive any internal cash flows to alleviate capital market frictions. Therefore, their cost parameter remains strictly higher than that of the PDL winners and falls within the interval $k \in \big(k(V_c^D), k(0)\big)$. Once combined with rebate disincentives, this relative cost disadvantage inhibits future innovation.
- **Medicaid Market Access:** These are firms that clear both the clinical and pricing stages and capture the market M_c by obtaining monopoly status or win the duopoly auction. Specifically, they benefit from both channels of state-dependent innovation: they accumulate regulatory know-how while also realising a strictly positive internal cash flow from the Medicaid market. Consequently, their future marginal cost of future R&D effort lowers to either $k(V_c^M)$ or $k(V_c^D)$, thereby allowing them to overcome market disincentives.

Mapping to RQ3 The triple-difference specification in RQ3 estimates class-specific responses to Medicaid expansion:

$$Y_{f,c,t} = \alpha_f + \delta_c + \lambda_t + \sum_{a \in \mathcal{A}} \theta_a \left(\mathbf{1}\{t \geq 2014\} \times \text{MedicaidShare}_c \times \mathbf{1}\{c = a\}\right) + \varepsilon_{f,c,t} \quad (55)$$

θ_a specifically estimates the innovation response in class a relative to that class's Medicaid exposure. The sign of θ_c is split into three categories as listed in Table 12.

Category	Sign of θ_a	Condition	Prediction
Volume-dominated	$\theta_a > 0$	$M_c^{post} < M_c^*$	$de_c^*/d\Delta M_c > 0$
Near-threshold	$\theta_a \approx 0$	$M_c^{post} \approx M_c^*$	$de_c^*/d\Delta M_c \approx 0$
Trap-bound	$\theta_a < 0$	$M_c^{post} > M_c^*$	$de_c^*/d\Delta M_c < 0$

Table 12: Mapping between theoretical categories and the RQ3 coefficient θ_a

Positive coefficients are consistent with the standard market-size effect, negative coefficients indicate classes where rebate extraction dominates the increased demand and insignificant coefficients suggest that the volume and monopsony effects balance each other.

9 Policy Implications

The model and empirical results identify a key trade-off in public pharmaceutical procurement. Expanding the Medicaid market strengthens states' bargaining position, allowing them to negotiate deeper supplemental rebates and reduce per-prescription drug expenditure. However, when rebate extraction lowers profits in the monopoly state sufficiently, the returns to successful innovation fall and incentives to invest in R&D decline. The model formalises this through the rebate function $\tau_c(M_c)$ and the resulting effect on V_c^M. Importantly, this does not imply that public demand expansion inherently harms innovation. Lakdawalla and Sood (2009) show that government premium subsidies can raise innovator profits under expansions of insured demand, due to profit margins being unaffected. The Medicaid setting is institutionally different - the expected rise in revenues generated by eligibility expansions are extracted through statutory rebates, supplemental rebate negotiations, and bargaining on the PDL, compressing the net price received by innovators rather than preserving it. When rebate extraction is sufficiently steep, expanding Medicaid's market size lowers the value of the monopoly state, resulting in a failure of the standard market-size prediction in this setting.

This trade-off is concentrated in a subset of therapeutic areas rather than uniformly applying across the pharmaceutical industry. In most therapeutic classes, the volume effect either dominates or offsets the monopsony effect, so that the net innovation response to the ACA Medicaid expansion is positive or negligible. Importantly, the incentive trap only binds in classes where the post-expansion market size exceeds the class-specific threshold M_c^*. The model predicts this condition is most likely to be met in classes with high drug substitutability, which lowers M_c^* by steepening the rebate schedule (Results 1), and high Medicaid exposure which raises M_c^{post} (Result 2) - consistent with the empirical findings. Consequently, a blanket policy response - such as uniform supplemental rebate caps across all drug classes - would be poorly targeted. It would reduce fiscal savings in classes where innovation has not been discouraged, while it may not necessarily address stalled innovation in classes where the rent gap has been most compressed.

Based on the comparative statics, the lowering of the rent gap $\Delta V_c = V_c^M - V_c^D$ is a sufficient statistic for reduced innovation efforts. Regardless of the functional form of the rebate schedule τ_c, any policy that compresses the monopoly premium also reduces equilibrium effort now and raises the future cost of innovation through the internal-finance channel. For this reason, the relevant policy question is not about the Medicaid market size, but about how Medicaid shapes the gap in payoffs in the monopoly and competitive states.

The model points towards two routes by which policy could mitigate this trade-off. First, policies that limit the rate at which rebates grow with market size, or that treat novel drugs differently from incumbent substitutes, can raise the threshold M_c^*, making the incentive trap less likely to bind, while at the cost of reducing Medicaid's fiscal savings from procurement. Second, R&D subsidies or other instruments lowering the effective cost of innovation k can maintain incentives in affected classes even while rebate pressure is strong. The balance between these routes therefore varies by class, depending on how far M_c^{post} exceeds M_c^*, and how narrow the rent gap has become.

The model moves beyond treating the savings-innovation trade-off as an industry-wide tension by identifying the rent gap ΔV_c and the market size threshold M_c^* as objects determining whether Medicaid's purchasing power discourages innovation within a given therapeutic area. This speaks to the broader policy debate, where Garthwaite and Starc (2023) argue that the central challenge of U.S. drug pricing reforms is balancing current affordability against the incentives driving investment into new therapies, and as discussed by Woods et al. (2024) in terms of dynamic efficiency. The results of this thesis suggest that this balance cannot be evaluated at the industry level, instead it must be assessed through the lens of the rent gap and market size threshold on a class by class basis.

10 Directions for Further Research

The empirical and theoretical analysis presented in this thesis is subject to several limitations. This section discusses the most important of these and identifies directions in which future work could extend the analysis.

10.1 Clinical Trial Outcomes

This thesis chooses to study patenting and drug approvals to capture the innovation stages before and after the clinical trials studied by Garthwaite et al. (2021). Clinical trials occupy an intermediate position between these two outcomes; incorporating clinical-trial data would complete the pipeline comparison, thereby allowing the analysis to trace the innovation response across all three stages. Given the scope of this study, this extension was not feasible. Garthwaite et al. utilise proprietary clinical trial data supplied by Cortellis. Constructing a comparable panel from ClinicalTrials.gov would require substantial work. The wording of study titles vary substantially and do not always identify the relevant disease area explicitly. Mapping these records to therapeutic-classes would therefore require imperfect text-based classification. Future work utilising Cortellis could assess whether the negative patenting response identified in this thesis is mirrored at the trial stage.

10.2 PDL Measurement

The PDL exposure measure used in this thesis is constructed from the Texas Vendor Drug Program, as this is the only state that has made the full formulary history available in a machine-readable format. While the representativeness analysis indicates that the Texas PDL shares substantial overlap with the New York and Washington lists, overlap with California is considerably weaker, reflecting differences in state-level formulary structures. This introduces measurement error into the PDL exposure variable, since firms classified as having heavy exposure to the PDL in Texas may be less exposed to it in other states. Consequently, the estimated innovation response to PDL exposure is likely to be attenuated towards zero, and should be interpreted as a plausible lower bound. Further work could construct a multi-state PDL exposure measure using additional formulary records, and average these across states to obtain a more representative measure of a firm's overall exposure to Medicaid's PDL.

10.3 Therapeutic Classification

The analysis aggregates all outcomes to a coarse therapeutic classification - ATC Level 1. Each class can contain drugs with substantially different clinical profiles and competitive environments, so within-class innovation heterogeneity may be masked. This is particularly relevant for the heterogeneity analysis in RQ3, where differences in innovation responses may remain hidden even where the ATC Level 1 estimates appear null or weak. Finer classification such as ATC Levels 2 or 3 may allow the analysis to distinguish between sub-therapeutic areas, but would come at the cost of an increase in panel dimensionality and therefore computational demands. Patents are classified using a CPC-to-ATC crosswalk, and extending to more granular ATC levels would require a mapping that may be noisy.

10.4 Alternative Estimators for the Joint Specification

The FE Poisson, binned-treatment, and continuous DiD estimators are applied to the RQ1 specification but not the RQ2 specification separating the innovation responses to both Medicaid and PDL exposure. Extending these estimators to the joint specification exceeded the computational power of the device used for this thesis, due to the size of the firm-class panels and the additional interactions required. The RQ2 estimates are instead supported by the ex-ante and ex-post exposure comparison with the use of FEOLS, but were not subject to equivalent estimator-level sensitivity analysis. Future work could therefore use greater computational capacity to assess whether the decomposition of demand and PDL exposure is robust to alternative models.

10.5 The Theoretical Model

The theoretical model developed is an illustrative simplification designed to formalise how the Medicaid expansion can reduce innovation incentives through the procurement design. The model restricts competition to two firms competing for access within each therapeutic class, and assumes that the winner of the price stage receives the entire Medicaid market. In reality, several manufacturers compete for PDL authorisation within a therapeutic class, and multiple products may receive preferred placement. The model therefore abstracts from an oligopolistic structure to a duopoly setting in order to isolate the interaction between market size, rebate extraction, and the rent gap rather than the exact number of firms. An N-firm extension would allow for the examination of how rebate pressure scales with the number of clinically viable competitors, and whether the Medicaid expansion causes marginal firms to exit Medicaid-relevant therapeutic areas entirely rather than simply reducing innovation effort.

11 Conclusion

This thesis has examined whether the ACA Medicaid expansion raised pharmaceutical innovation, and whether the response depended on firms' exposure to Medicaid's procurement institutions. Conventionally, an expansion in expected demand should increase the expected return to R&D (Finkelstein, 2004; Dubois et al., 2015). However, the Medicaid setting alters this prediction, as the additional demand is subject to statutory rebates, supplemental rebate negotiations, and drug list bargaining. To study this question, the analysis constructed novel firm-therapeutic-class panels linking both patent records and FDA drug approvals to Medicaid prescription data, and PDL placement over 2010-2025.

The main empirical finding is that therapeutic classes with greater exposure to Medicaid experienced a persistent decline in patenting following the 2014 reform, while FDA approvals had an inconclusive response. The divergence between patents and FDA approvals is consistent with the changes in patenting behaviour appearing earlier in the R&D pipeline, with insufficient time passing for changes in drug development to reach the approval stage. This interpretation is consistent with the insignificant effect documented in clinical trials by Garthwaite et al. (2021).

A decomposition of innovation behaviour by demand and PDL exposure revealed that firms with greater exposure to Medicaid's PDL experienced additional declines in patenting beyond the demand channel. The PDL counterfactual analysis provides further support for this. The aggregate decline in patenting is not uniform across the pharmaceutical industry, with the declines being concentrated in classes with higher exposure to Medicaid and greater levels of drug substitutability. However, classes with lower drug substitutability instead saw innovation responses consistent with the standard market-size prediction, with a rise in patent applications. The PDL counterfactual adds further to this nuanced mechanism, where firms passing only one stage of the PDL selection process experience the largest declines in patenting, while firms securing full preferred placement exhibited declines, albeit smaller ones.

The theoretical model formalises this procurement-based interpretation. Since rebate extraction strengthened with the Medicaid market expansion, additional demand can lower expected returns to R&D. This effect is likelier to hold for more clinically substitutable drugs, where Medicaid's bargaining position for deeper rebates is stronger. Notably, the model identifies the gap between monopoly and competitive profits as the key object linking pricing pressure to innovation incentives. Policies that shrink this gap therefore reduce equilibrium effort immediately and in the future period through an internal-financing channel.

Taken together, the findings of this thesis suggest that the consequences of public demand expansions on innovation depend on the interaction between institutions governing how firms capture surpluses and market conditions, such as within-therapeutic-class drug substitutability. The findings speak more broadly to reforms that leverage government purchasing power to place downward pressure on drug prices, including debates surrounding Medicare price negotiation under the 2022 Inflation Reduction Act. Crucially, the analysis suggests that the innovation consequences of such reforms should be evaluated on a class-by-class basis, since the same demand shock may encourage R&D in some therapeutic areas while weakening incentives in others.

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13 Appendix

13.1 Patent Assignee Matching

Table 13: Pre-2014 patent share: matched vs unmatched assignees

	Matched	Unmatched
Pre-2014 patent share	35.8%	34.6%
Post-2014 patent share	64.2%	65.4%

Table 14: ATC1 share of A61P patents: matched vs unmatched

ATC1	Matched	Unmatched
A	10.1%	10.3%
B	4.0%	3.8%
C	7.9%	7.7%
D	5.7%	5.7%
G	5.1%	4.5%
H	2.2%	1.8%
J	8.4%	9.4%
L	15.6%	16.6%
M	9.3%	9.6%
N	9.7%	9.7%
P	1.2%	1.4%
R	4.8%	4.4%
S	3.9%	3.3%
V	12.0%	11.7%

13.2 PDL Exposure Distribution

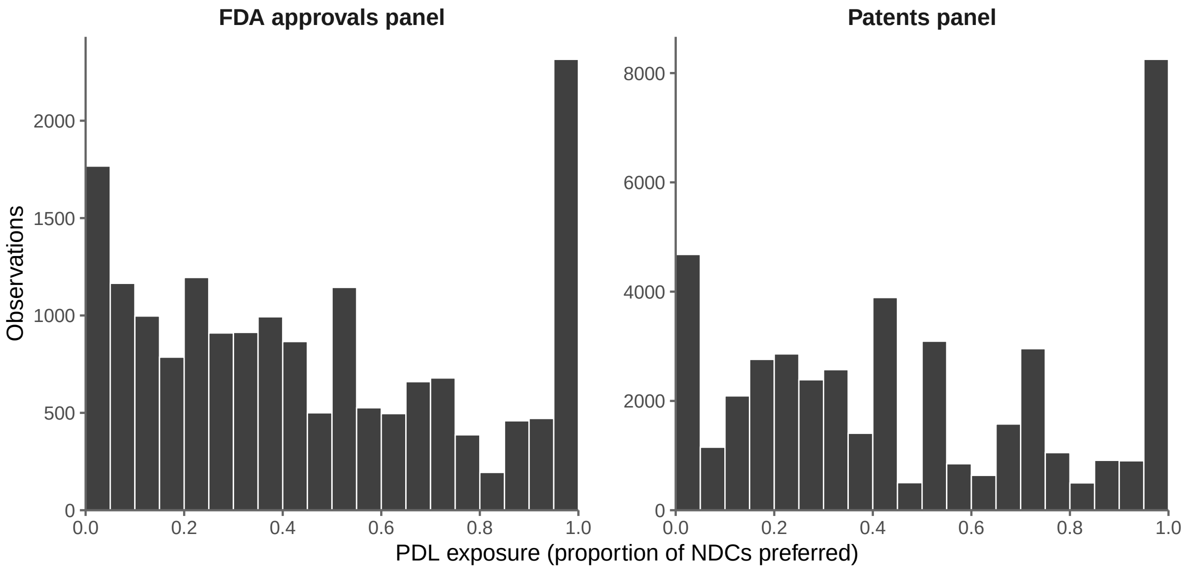

Figure 9: Distribution of PDL exposure in constructed panels

13.3 Regression Tables

13.3.1 RQ1: Overall Innovation Response

	Patents	FDA Approvals
2010 × Medicaid Share	−0.041	0.215
	(0.038)	(0.463)
2011 × Medicaid Share	−0.017	−0.017
	(0.036)	(0.381)
2012 × Medicaid Share	0.002	0.208
	(0.029)	(0.412)
2014 × Medicaid Share	−0.022	−0.321
	(0.027)	(0.463)
2015 × Medicaid Share	−0.086**	−0.053
	(0.030)	(0.477)
2016 × Medicaid Share	−0.125***	0.414
	(0.034)	(0.423)
2017 × Medicaid Share	−0.114**	1.421*
	(0.036)	(0.677)
2018 × Medicaid Share	−0.132***	0.999
	(0.039)	(0.550)
2019 × Medicaid Share	−0.189***	0.554
	(0.041)	(0.538)
2020 × Medicaid Share	−0.220***	0.732
	(0.043)	(0.509)
2021 × Medicaid Share	−0.220***	0.137
	(0.043)	(0.632)
2022 × Medicaid Share	−0.233***	0.365
	(0.043)	(0.645)
2023 × Medicaid Share	−0.243***	0.846
	(0.044)	(0.641)
2024 × Medicaid Share	−0.252***	0.258
	(0.046)	(0.523)
2025 × Medicaid Share	−0.228***	−0.137
	(0.044)	(0.510)
Num. obs.	5 281 696	151 200
R2	0.534	0.609

* p <0.05, ** p <0.01, *** p <0.001

Table 15: FEOLS event-study estimates: overall innovation response (ex-ante)

13.3.2 RQ2: Medicaid Demand and PDL Exposure

	Patents		FDA Approvals	
	Medicaid Market Share	PDL Exposure	Medicaid Market Share	PDL Exposure
2010	−0.041	1.546	0.214	0.049
	(0.038)	(2.018)	(0.461)	(0.807)
2011	−0.017	0.734	−0.010	−0.243
	(0.036)	(1.642)	(0.382)	(0.712)
2012	0.002	1.079	0.171	1.402
	(0.029)	(1.289)	(0.406)	(0.911)
2014	−0.022	4.161*	−0.333	0.403
	(0.026)	(1.890)	(0.465)	(0.647)
2015	−0.086**	0.836	−0.088	1.163
	(0.030)	(1.480)	(0.478)	(0.881)
2016	−0.126***	−1.554	0.334	2.645
	(0.034)	(1.620)	(0.426)	(1.683)
2017	−0.114**	−1.342	1.364*	1.923
	(0.036)	(1.890)	(0.680)	(0.994)
2018	−0.132***	−2.786	0.949	1.684
	(0.039)	(1.644)	(0.552)	(1.021)
2019	−0.189***	−3.806*	0.519	1.164*
	(0.041)	(1.912)	(0.539)	(0.592)
2020	−0.220***	−4.649*	0.687	1.579
	(0.043)	(1.894)	(0.514)	(0.942)
2021	−0.220***	−5.266**	0.038	3.114*
	(0.043)	(1.687)	(0.640)	(1.394)
2022	−0.233***	−6.029**	0.333	0.994
	(0.044)	(1.941)	(0.648)	(0.655)
2023	−0.243***	−6.925***	0.809	1.185
	(0.044)	(1.851)	(0.649)	(1.738)
2024	−0.252***	−6.484***	0.272	−0.452
	(0.046)	(1.788)	(0.523)	(0.590)
2025	−0.227***	−4.588**	−0.109	−1.118
	(0.044)	(1.549)	(0.506)	(0.908)
Num. obs.	5 281 696		151 200	
R2	0.535		0.609	

* p <0.05, ** p <0.01, *** p <0.001

Table 16: FEOLS event-study estimates: Medicaid demand and PDL exposure (ex-ante)

13.4 Therapeutic-Class Substitutability

Table 17: Average annual distinct NDC count by ATC class

ATC	Therapeutic class	Mean Annual NDCs
N	Nervous system	7351
C	Cardiovascular system	4297
A	Alimentary tract and metabolism	3407
J	Anti-infectives for systemic use	2332
D	Dermatologicals	1771
R	Respiratory system	1672
M	Musculo-skeletal system	1407
B	Blood and blood-forming organs	1389
L	Antineoplastic and immunomodulating	932
G	Genito-urinary and sex hormones	906
H	Systemic hormonal preparations	768
S	Sensory organs	579
V	Various	211
P	Antiparasitic products	165

13.5 Robustness Regression Tables

13.5.1 RQ1: Ex-Post FEOLS

	Patents	FDA Approvals
2010 × Medicaid Share	−0.033	0.286
	(0.035)	(0.450)
2011 × Medicaid Share	−0.010	0.336
	(0.033)	(0.413)
2012 × Medicaid Share	0.004	0.414
	(0.027)	(0.455)
2014 × Medicaid Share	−0.018	−0.324
	(0.024)	(0.533)
2015 × Medicaid Share	−0.088**	0.005
	(0.027)	(0.420)
2016 × Medicaid Share	−0.123***	0.550
	(0.032)	(0.413)
2017 × Medicaid Share	−0.122***	1.635*
	(0.034)	(0.688)
2018 × Medicaid Share	−0.135***	1.240*
	(0.037)	(0.560)
2019 × Medicaid Share	−0.194***	0.400
	(0.039)	(0.497)
2020 × Medicaid Share	−0.225***	0.652
	(0.041)	(0.529)
2021 × Medicaid Share	−0.222***	0.218
	(0.040)	(0.573)
2022 × Medicaid Share	−0.233***	0.418
	(0.040)	(0.590)
2023 × Medicaid Share	−0.238***	0.842
	(0.040)	(0.576)
2024 × Medicaid Share	−0.250***	0.137
	(0.042)	(0.502)
2025 × Medicaid Share	−0.212***	−0.380
	(0.041)	(0.531)
Num. obs.	5 281 696	151 200
R2	0.534	0.609

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 18: FEOLS event-study estimates: overall innovation response (ex-post)

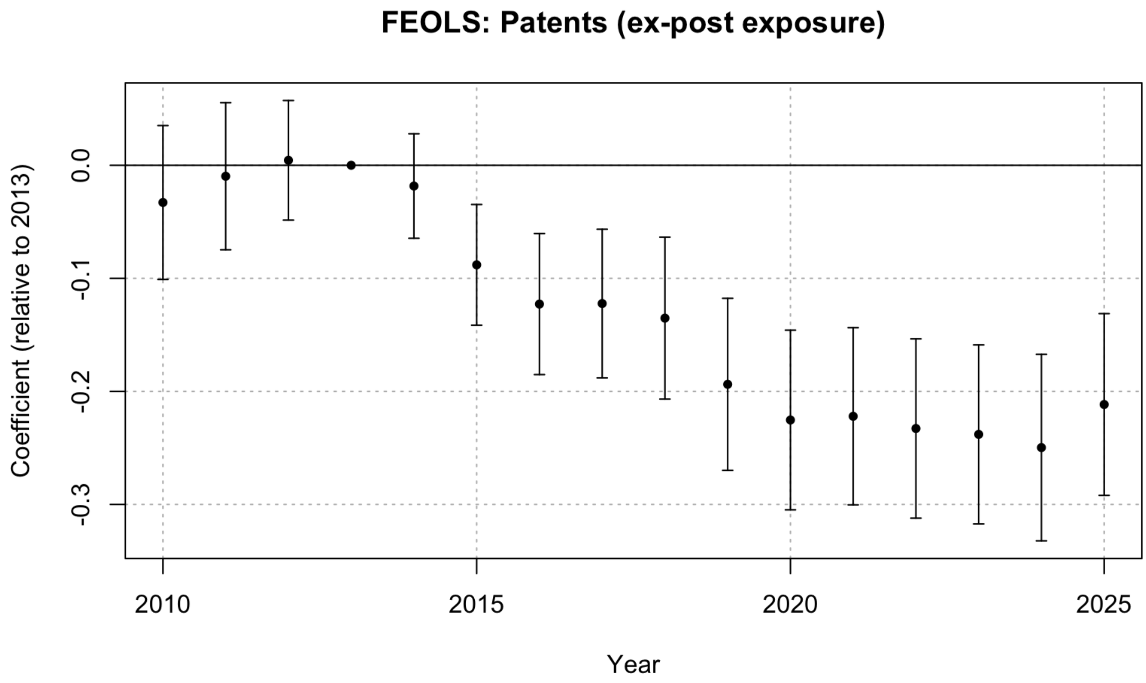

Figure 10: FEOLS event study: patents, ex-post Medicaid exposure

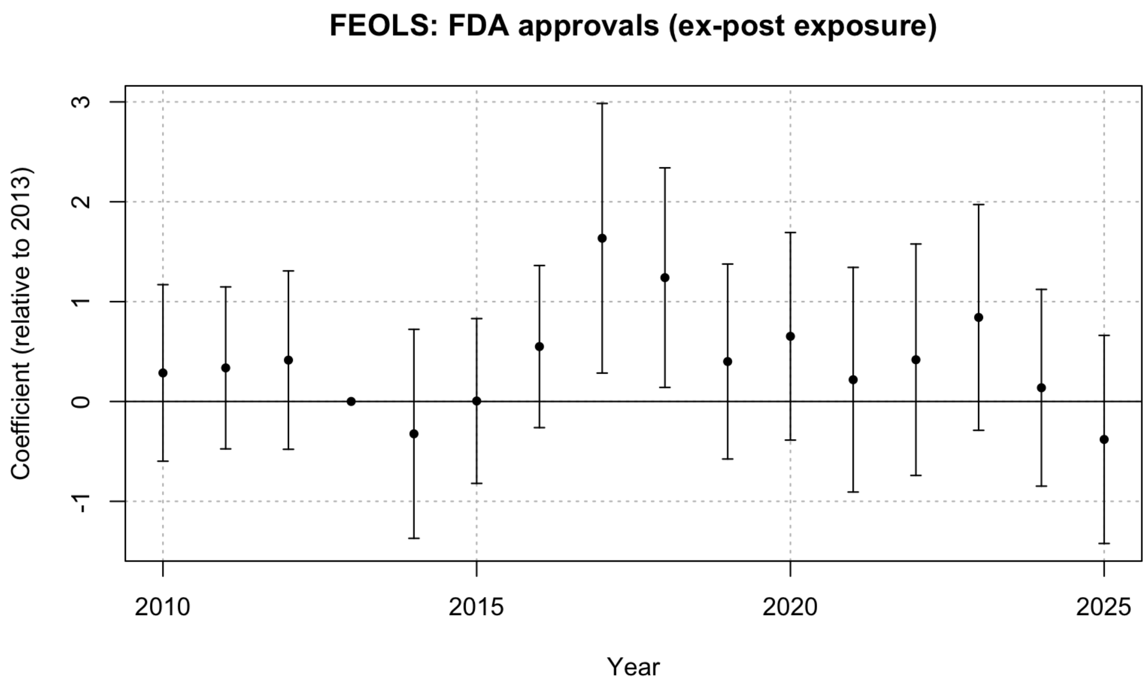

Figure 11: FEOLS event study: FDA approvals, ex-post Medicaid exposure

13.5.2 RQ1: Ex-Ante FE Poisson

	Patents	FDA Approvals
2010 × Medicaid Share	0.120	0.237
	(0.154)	(0.636)
2011 × Medicaid Share	0.218	0.326
	(0.140)	(0.446)
2012 × Medicaid Share	0.115	0.643
	(0.108)	(0.615)
2014 × Medicaid Share	−0.092	−0.545
	(0.108)	(0.638)
2015 × Medicaid Share	−0.300*	−0.434
	(0.118)	(0.632)
2016 × Medicaid Share	−0.435***	−0.834
	(0.124)	(0.478)
2017 × Medicaid Share	−0.397**	−0.495
	(0.131)	(0.480)
2018 × Medicaid Share	−0.415**	−0.772
	(0.149)	(0.530)
2019 × Medicaid Share	−0.784***	−0.381
	(0.154)	(0.565)
2020 × Medicaid Share	−0.998***	−0.733
	(0.173)	(0.574)
2021 × Medicaid Share	−0.989***	−0.867
	(0.170)	(0.729)
2022 × Medicaid Share	−1.137***	−0.637
	(0.186)	(0.594)
2023 × Medicaid Share	−1.324***	−0.566
	(0.201)	(0.511)
2024 × Medicaid Share	−1.474***	−0.680
	(0.261)	(0.573)
2025 × Medicaid Share	−1.110***	−0.668
	(0.229)	(0.580)
Num. obs.	1 300 944	42 176

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 19: FE Poisson event-study estimates: overall innovation response (ex-ante)

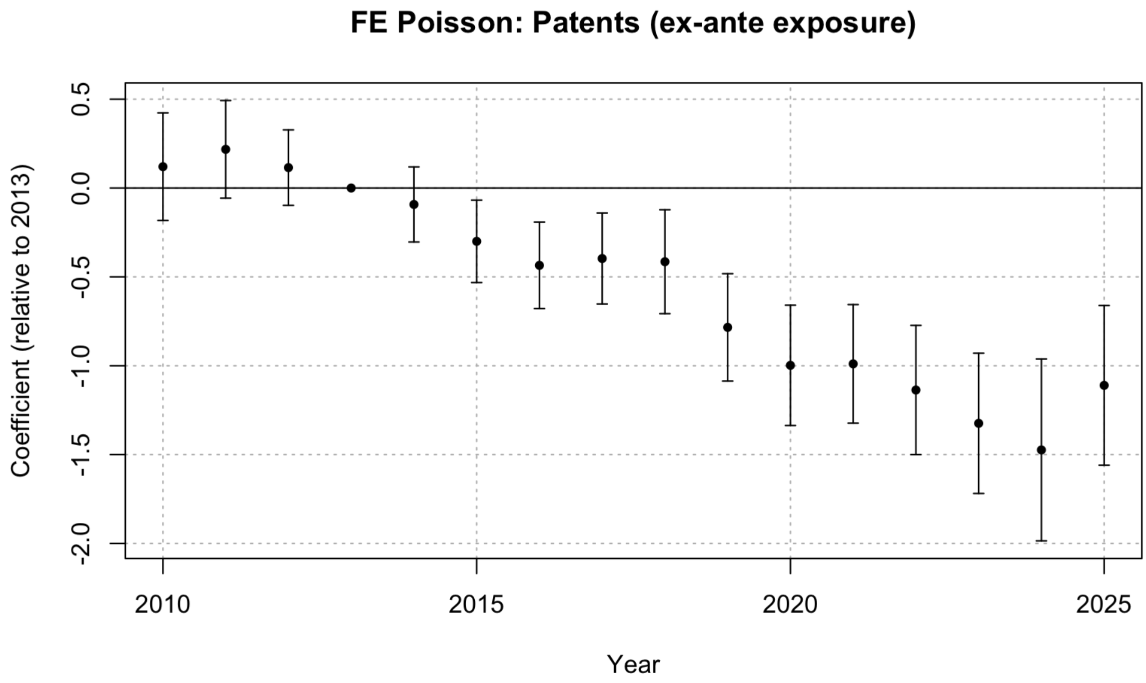

Figure 12: FE Poisson event study: patents, ex-ante Medicaid exposure

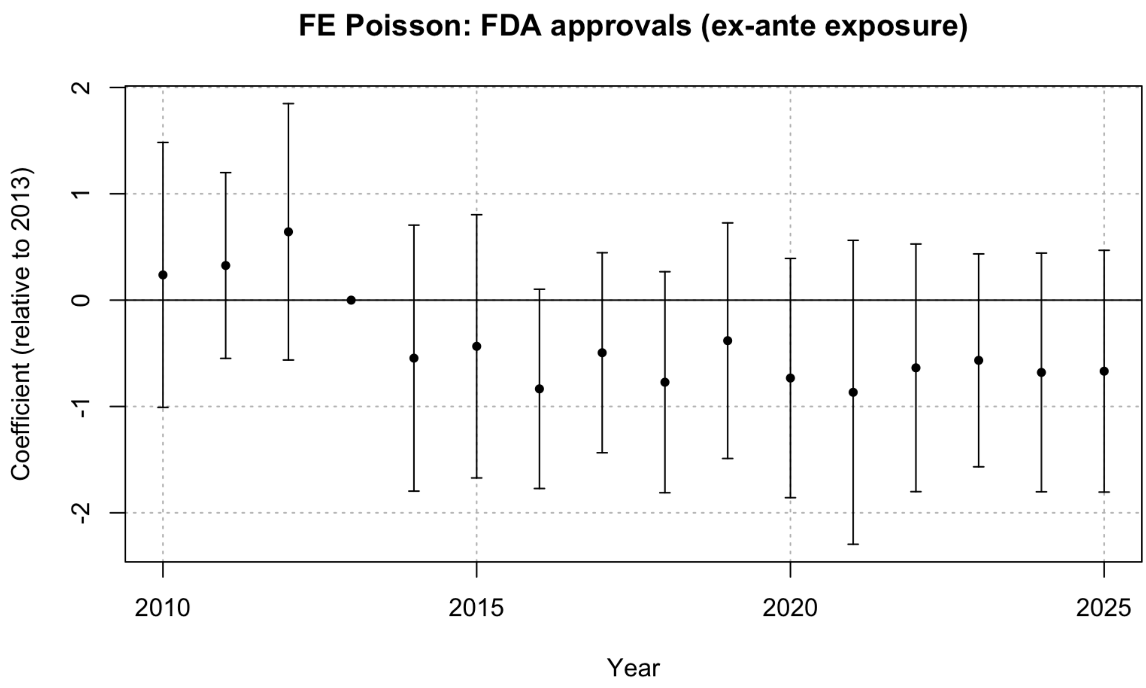

Figure 13: FE Poisson event study: FDA approvals, ex-ante Medicaid exposure

13.5.3 RQ1: Ex-Post FE Poisson

	Patents	FDA Approvals
2010 × Medicaid Share	0.129	0.453
	(0.157)	(0.869)
2011 × Medicaid Share	0.240	1.138
	(0.145)	(0.687)
2012 × Medicaid Share	0.124	1.250
	(0.115)	(0.863)
2014 × Medicaid Share	−0.086	−0.711
	(0.106)	(0.996)
2015 × Medicaid Share	−0.367**	−0.431
	(0.119)	(0.725)
2016 × Medicaid Share	−0.515***	−0.788
	(0.129)	(0.660)
2017 × Medicaid Share	−0.533***	−0.278
	(0.134)	(0.735)
2018 × Medicaid Share	−0.556***	−0.595
	(0.157)	(0.713)
2019 × Medicaid Share	−0.998***	−0.625
	(0.167)	(0.764)
2020 × Medicaid Share	−1.293***	−0.916
	(0.190)	(0.839)
2021 × Medicaid Share	−1.313***	−0.909
	(0.188)	(0.910)
2022 × Medicaid Share	−1.564***	−0.657
	(0.203)	(0.775)
2023 × Medicaid Share	−1.803***	−0.618
	(0.224)	(0.695)
2024 × Medicaid Share	−2.021***	−0.969
	(0.266)	(0.854)
2025 × Medicaid Share	−1.692***	−1.254
	(0.262)	(0.892)
Num. obs.	1 300 944	42 176

* p <0.05, ** p <0.01, *** p <0.001

Table 20: FE Poisson event-study estimates: overall innovation response (ex-post)

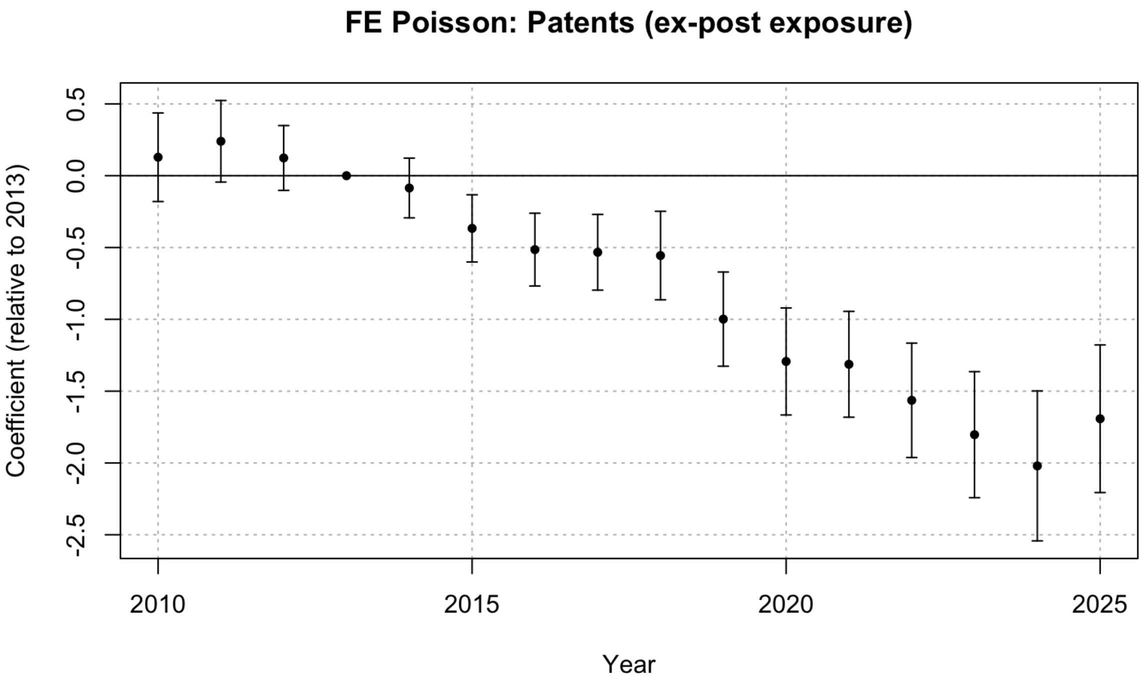

Figure 14: FE Poisson event study: patents, ex-post Medicaid exposure

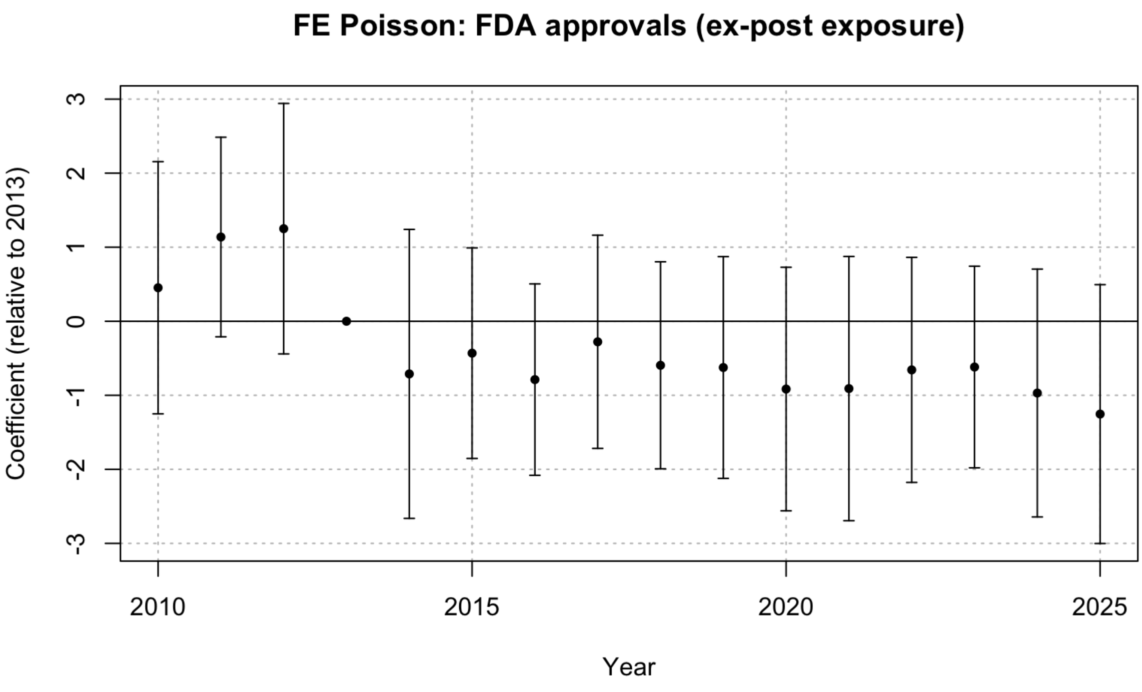

Figure 15: FE Poisson event study: FDA approvals, ex-post Medicaid exposure

13.5.4 RQ1: Binned Treatment

Table 21: Pre-expansion Medicaid market shares and bin assignments

ATC	Therapeutic Area	Medicaid Share	Bin
N	Nervous system	0.363	High
C	Cardiovascular system	0.215	High
J	Anti-infectives for systemic use	0.120	Medium
D	Dermatologicals	0.099	Medium
R	Respiratory system	0.085	Low
A	Alimentary tract and metabolism	0.068	Low
M	Musculo-skeletal system	0.015	Low
H	Systemic hormonal preparations	0.012	Low
L	Antineoplastic and immunomodulating agents	0.006	Low
G	Genito-urinary system and sex hormones	0.006	Low
S	Sensory organs	0.005	Low
B	Blood and blood-forming organs	0.005	Low
V	Various	0.000	Low
P	Antiparasitic products	0.000	Low

	Patents		FDA Approvals	
	Medium Bin	High Bin	Medium Bin	High Bin
2010	−0.008	−0.029	0.042	0.085
	(0.006)	(0.017)	(0.055)	(0.140)
2011	−0.005	−0.025	0.058	−0.064
	(0.006)	(0.016)	(0.065)	(0.105)
2012	0.009	−0.003	0.042	0.033
	(0.006)	(0.012)	(0.077)	(0.121)
2014	0.015**	−0.002	0.047	−0.091
	(0.005)	(0.011)	(0.067)	(0.130)
2015	0.002	−0.027*	0.006	−0.005
	(0.005)	(0.013)	(0.073)	(0.148)
2016	0.007	−0.058***	0.158	0.070
	(0.006)	(0.014)	(0.101)	(0.130)
2017	0.005	−0.047**	0.239*	0.326
	(0.006)	(0.015)	(0.114)	(0.181)
2018	−0.009	−0.075***	0.185*	0.213
	(0.006)	(0.017)	(0.093)	(0.155)
2019	−0.004	−0.092***	0.071	0.206
	(0.007)	(0.017)	(0.077)	(0.162)
2020	−0.003	−0.108***	0.012	0.141
	(0.006)	(0.018)	(0.092)	(0.142)
2021	−0.005	−0.118***	0.127	−0.048
	(0.006)	(0.018)	(0.105)	(0.182)
2022	−0.010	−0.129***	0.101	0.040
	(0.006)	(0.018)	(0.075)	(0.184)
2023	−0.011	−0.141***	0.152	0.154
	(0.006)	(0.019)	(0.087)	(0.182)
2024	−0.005	−0.142***	0.037	0.070
	(0.010)	(0.019)	(0.066)	(0.146)
2025	−0.020**	−0.155***	0.010	0.032
	(0.006)	(0.019)	(0.067)	(0.153)
Num. obs.	5 281 696		151 200	
R2	0.534		0.609	

* p <0.05, ** p <0.01, *** p <0.001

Table 22: Binned treatment event-study estimates (FEOLS, ex-ante)

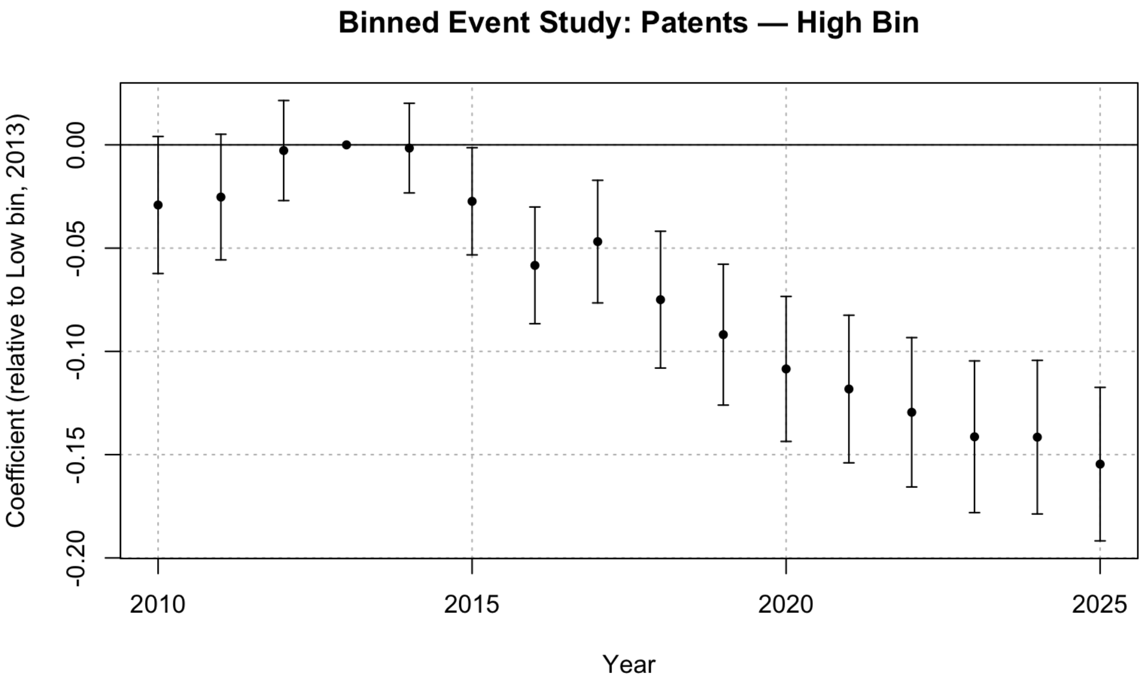

Figure 16: Binned event study: patents, high-exposure bin (N, C)

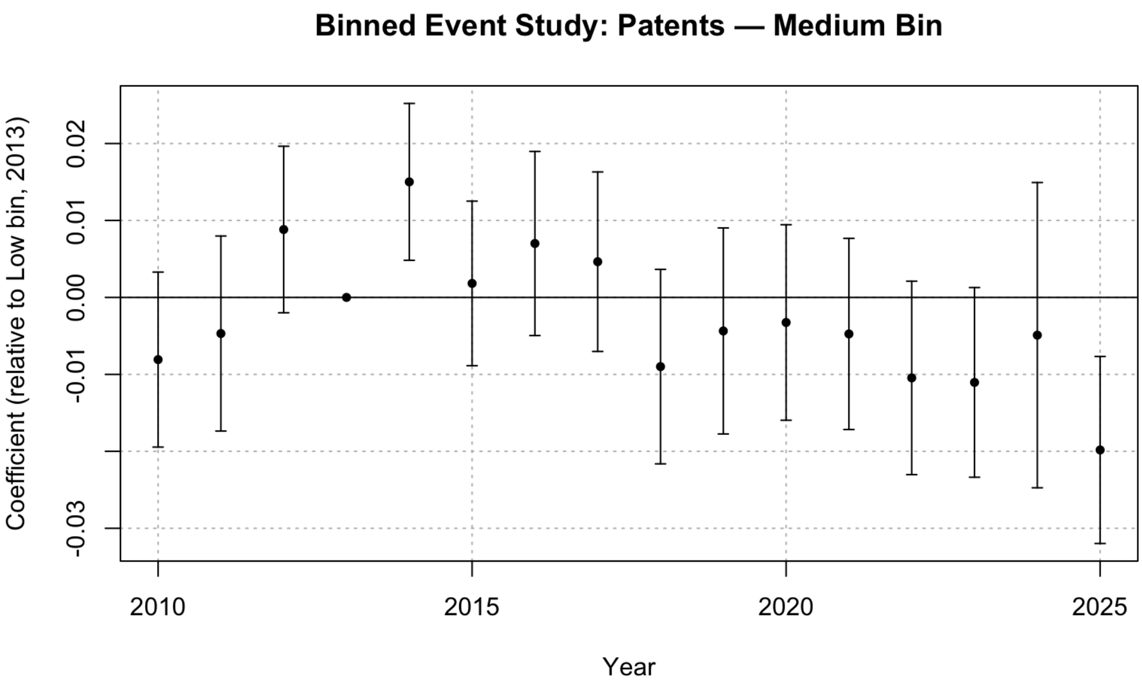

Figure 17: Binned event study: patents, medium-exposure bin (J, D)

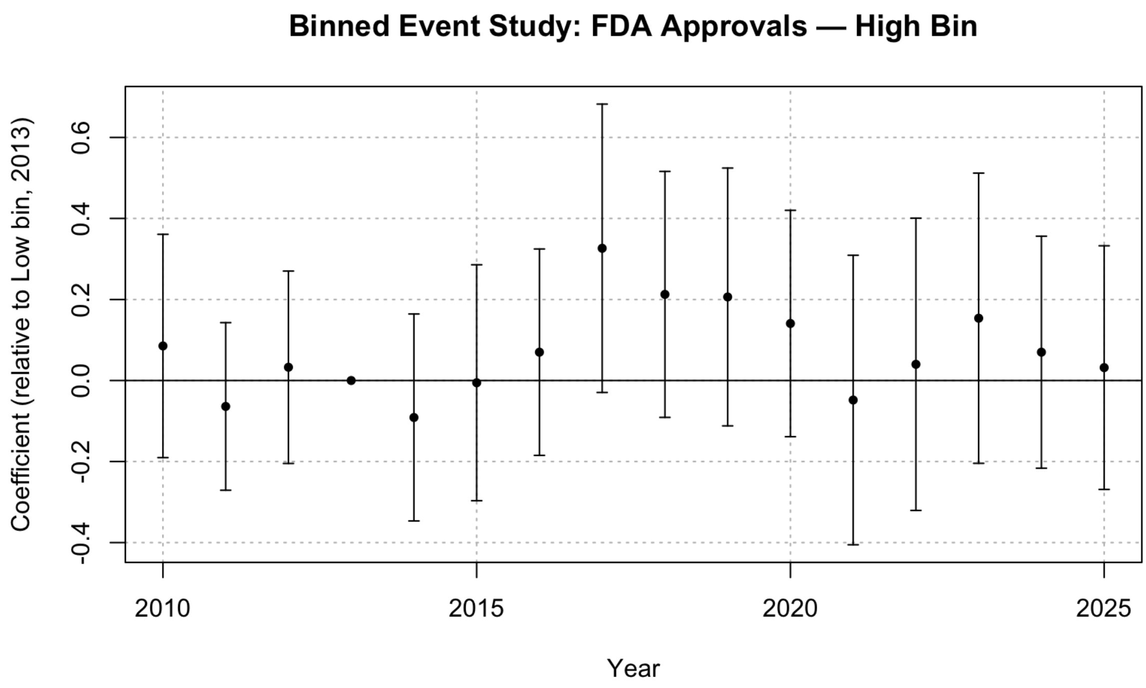

Figure 18: Binned event study: FDA approvals, high-exposure bin (N, C)

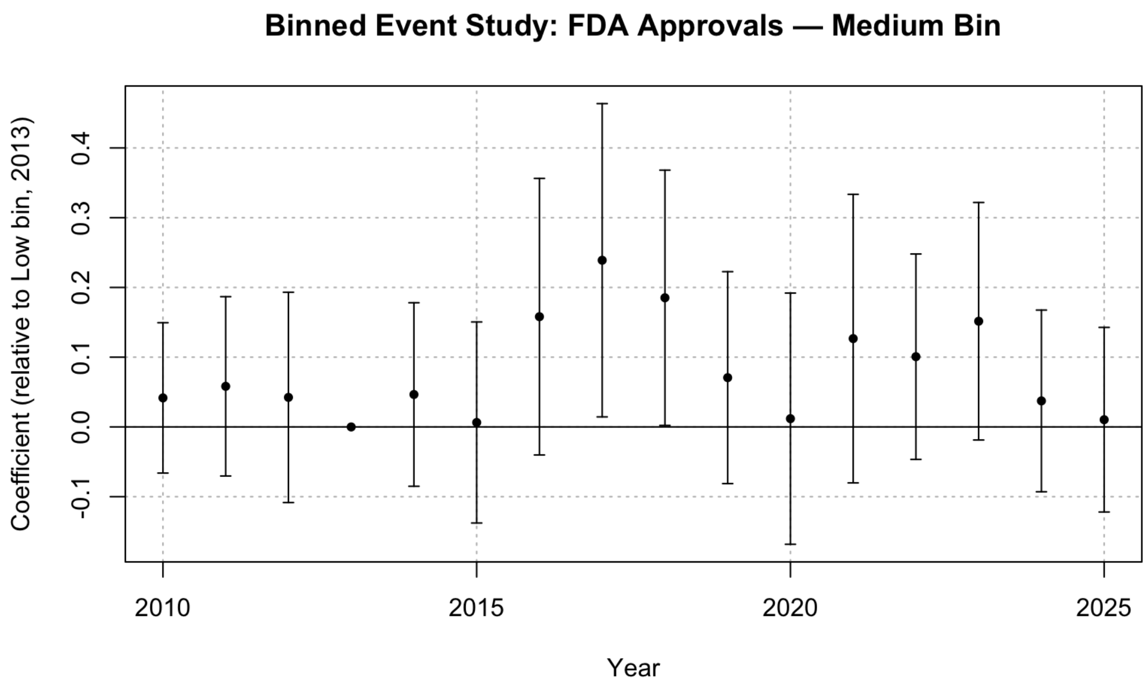

Figure 19: Binned event study: FDA approvals, medium-exposure bin (J, D)

13.5.5 RQ1: Continuous Difference-in-Differences

Table 23: Continuous DiD event-study estimates

	Patents	FDA Approvals
2011	0.0038	−0.0368
	(0.0034)	(0.0602)
2012	0.0023	0.0361
	(0.0037)	(0.0576)
2013	0.0002	−0.0326
	(0.0050)	(0.0632)
2014	−0.0025	−0.0506
	(0.0044)	(0.0731)
2015	−0.0126	−0.0092
	(0.0055)	(0.0706)
2016	−0.0181*	0.0646
	(0.0062)	(0.0721)
2017	−0.0165	0.2233
	(0.0069)	(0.1018)
2018	−0.0189	0.1568
	(0.0073)	(0.0930)
2019	−0.0271*	0.0861
	(0.0080)	(0.0866)
2020	−0.0313*	0.1151
	(0.0082)	(0.0846)
2021	−0.0315*	0.0223
	(0.0084)	(0.1022)
2022	−0.0333*	0.0579
	(0.0083)	(0.1086)
2023	−0.0348*	0.1329
	(0.0084)	(0.1056)
2024	−0.0358*	0.0404
	(0.0085)	(0.0921)
2025	−0.0324*	−0.0213
	(0.0091)	(0.0919)

* 95% uniform confidence band does not cover 0

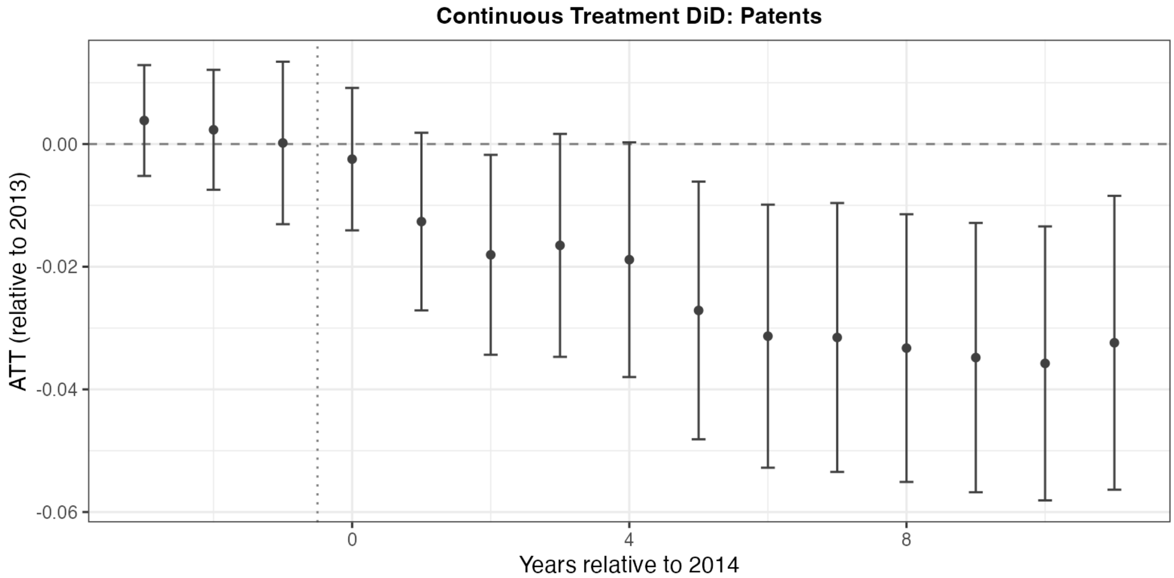

Figure 20: Continuous DiD event study: patents

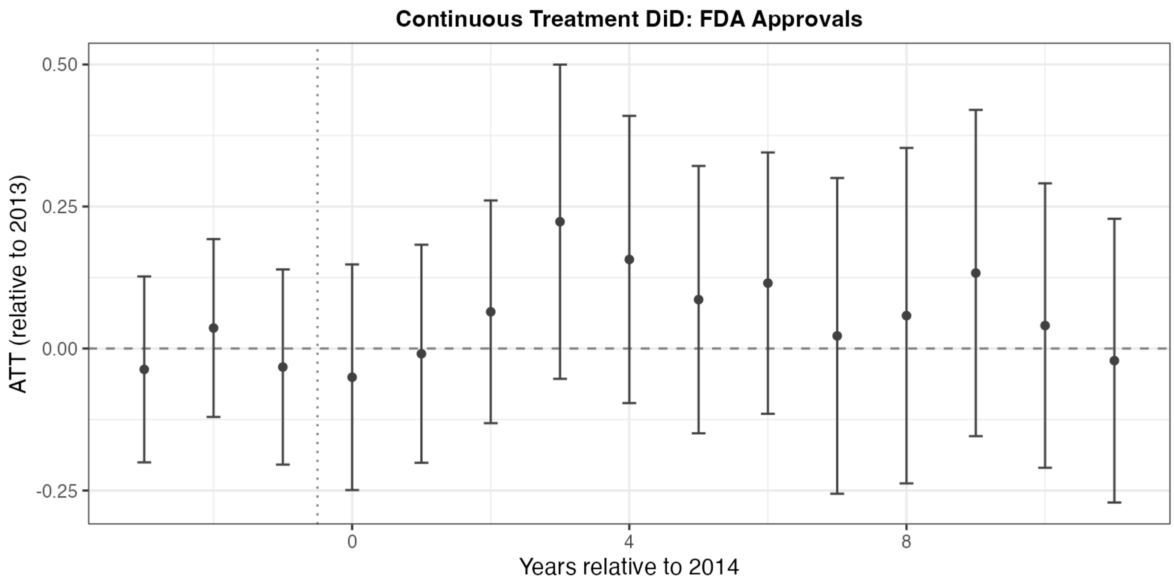

Figure 21: Continuous DiD event study: FDA approvals

13.5.6 RQ2: Ex-Post FEOLS

	Patents		FDA Approvals	
	Medicaid Market Share	PDL Exposure	Medicaid Market Share	PDL Exposure
2010	−0.033	1.546	0.285	0.046
	(0.035)	(2.018)	(0.448)	(0.807)
2011	−0.010	0.734	0.344	−0.257
	(0.033)	(1.642)	(0.412)	(0.711)
2012	0.005	1.079	0.371	1.395
	(0.027)	(1.289)	(0.447)	(0.912)
2014	−0.017	4.161*	−0.338	0.398
	(0.024)	(1.890)	(0.536)	(0.647)
2015	−0.088**	0.836	−0.037	1.158
	(0.027)	(1.480)	(0.422)	(0.881)
2016	−0.123***	−1.554	0.456	2.642
	(0.032)	(1.620)	(0.418)	(1.682)
2017	−0.123***	−1.342	1.566*	1.924
	(0.034)	(1.890)	(0.690)	(0.992)
2018	−0.136***	−2.787	1.181*	1.683
	(0.037)	(1.644)	(0.563)	(1.021)
2019	−0.195***	−3.807*	0.359	1.171*
	(0.039)	(1.912)	(0.498)	(0.591)
2020	−0.227***	−4.650*	0.598	1.585
	(0.041)	(1.894)	(0.537)	(0.942)
2021	−0.224***	−5.267**	0.103	3.110*
	(0.040)	(1.687)	(0.583)	(1.393)
2022	−0.235***	−6.029**	0.381	0.994
	(0.041)	(1.941)	(0.593)	(0.653)
2023	−0.240***	−6.926***	0.798	1.191
	(0.041)	(1.851)	(0.583)	(1.737)
2024	−0.252***	−6.485***	0.154	−0.447
	(0.042)	(1.788)	(0.501)	(0.589)
2025	−0.212***	−4.589**	−0.347	−1.113
	(0.041)	(1.549)	(0.525)	(0.908)
Num. obs.	5 281 696		151 200	
R2	0.535		0.609	

* p <0.05, ** p <0.01, *** p <0.001

Table 24: FEOLS event-study estimates: Medicaid demand and PDL exposure (ex-post)

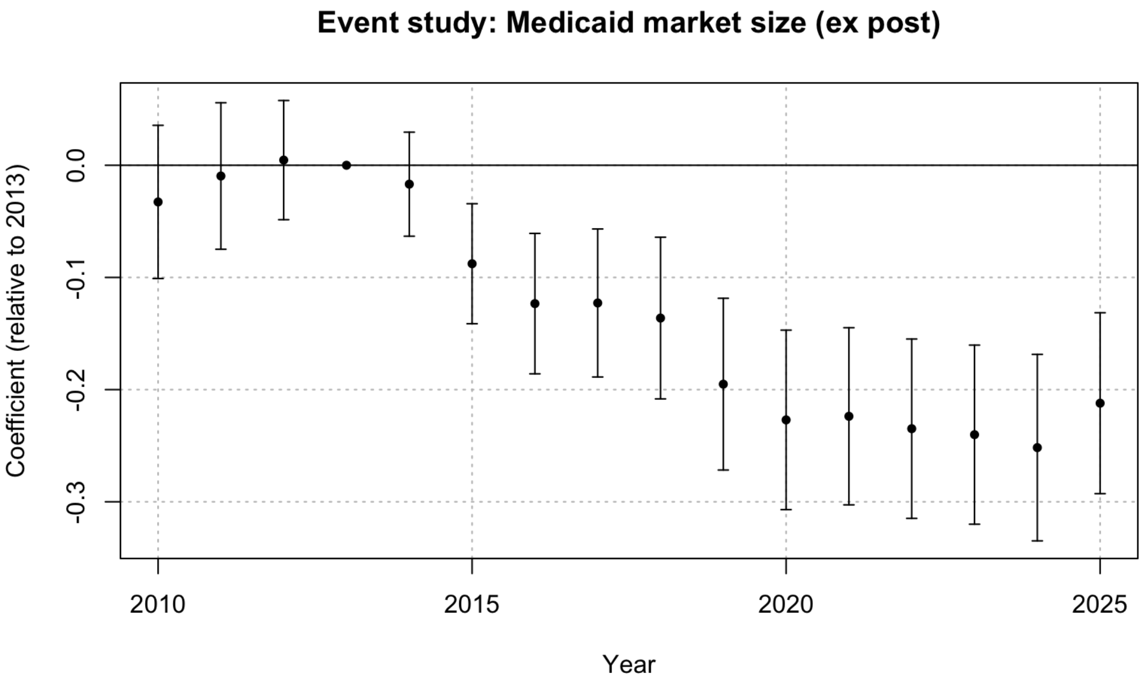

Figure 22: FEOLS event study: patents, Medicaid exposure (ex-post)

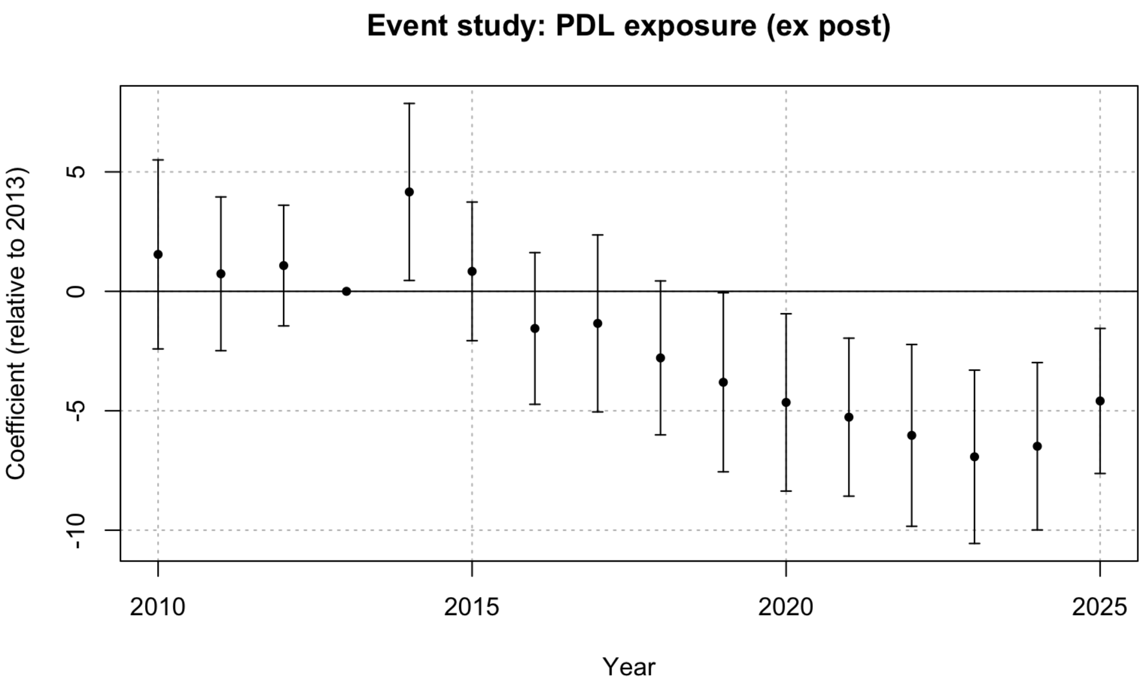

Figure 23: FEOLS event study: patents, PDL exposure (ex-post)

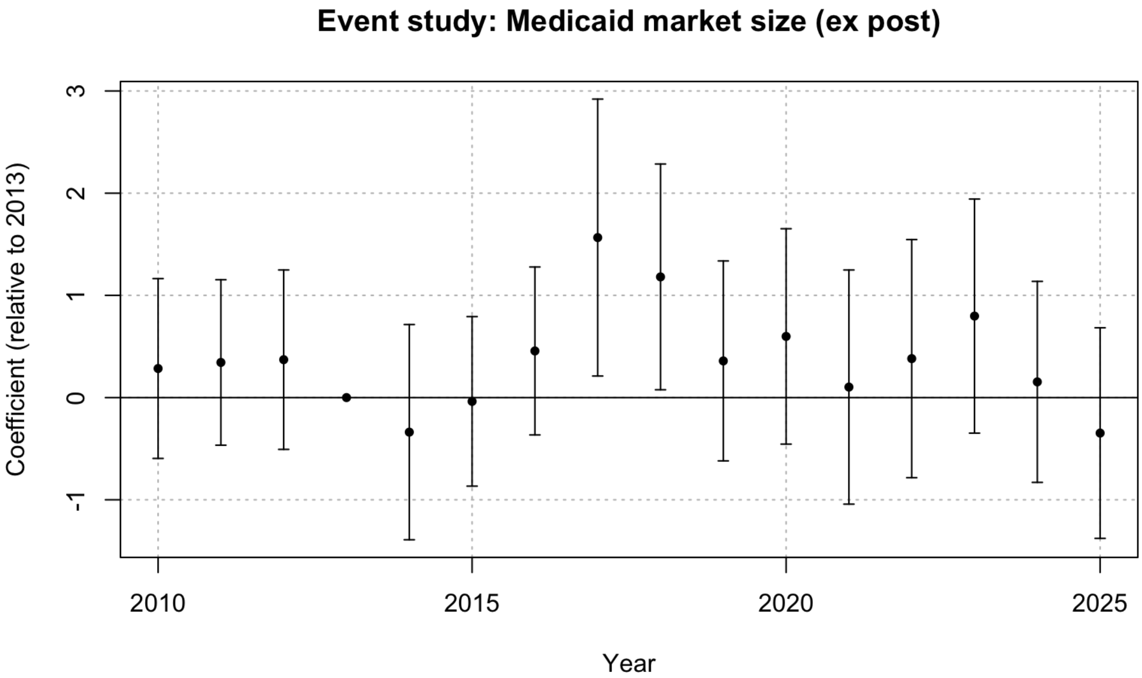

Figure 24: FEOLS event study: FDA approvals, Medicaid exposure (ex-post)

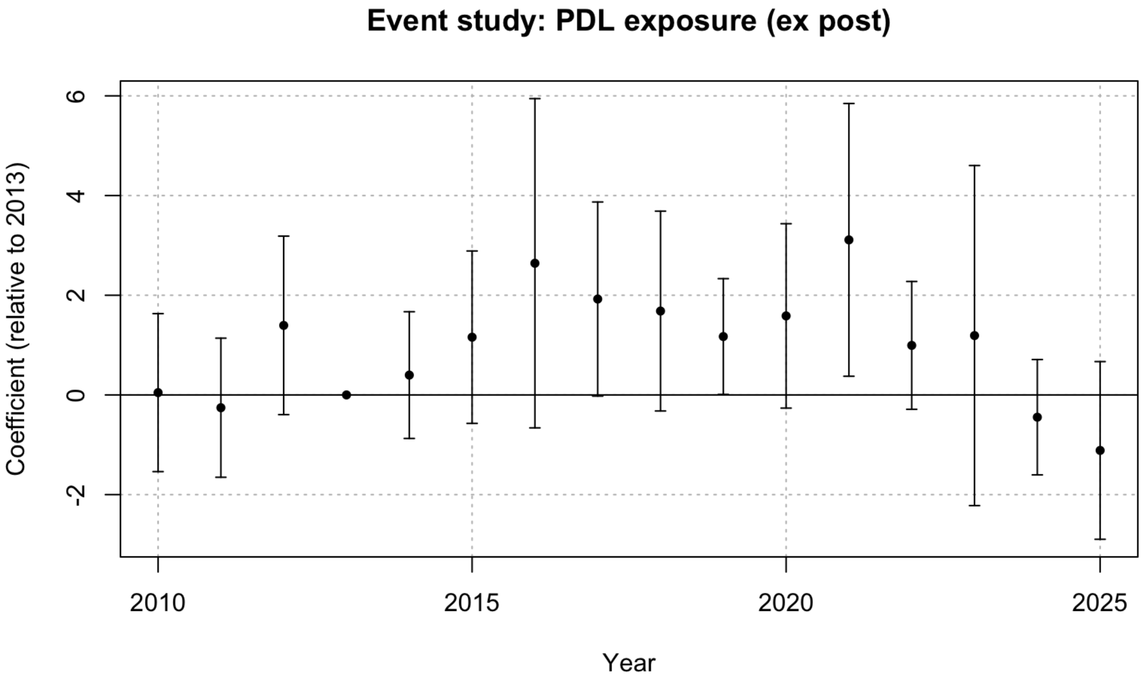

Figure 25: FEOLS event study: FDA approvals, PDL exposure (ex-post)

13.5.7 RQ3: Ex-Post FEOLS

	Patents	FDA Approvals
Medicaid Share × ATC = A	−0.157***	1.301**
	(0.031)	(0.420)
Medicaid Share × ATC = B	0.151***	−0.121
	(0.027)	(0.784)
Medicaid Share × ATC = C	−0.094***	0.373
	(0.015)	(0.278)
Medicaid Share × ATC = D	−0.017	0.881**
	(0.062)	(0.337)
Medicaid Share × ATC = G	0.032	−1.058
	(0.049)	(1.121)
Medicaid Share × ATC = H	0.619***	0.446
	(0.096)	(0.530)
Medicaid Share × ATC = J	0.035	−0.078
	(0.034)	(0.089)
Medicaid Share × ATC = L	0.443	−0.244
	(0.578)	(2.115)
Medicaid Share × ATC = M	−0.212***	0.017
	(0.048)	(0.858)
Medicaid Share × ATC = N	−0.174***	0.047
	(0.039)	(0.101)
Medicaid Share × ATC = R	0.005	0.178
	(0.003)	(0.130)
Medicaid Share × ATC = S	0.105***	6.473*
	(0.020)	(3.004)
Num. obs.	19 617 728	561 600
R2	0.247	0.235

* p <0.05, ** p <0.01, *** p <0.001

Table 25: Triple-difference estimates: therapeutic-class heterogeneity (FEOLS, ex-post)

13.6 Baseline Model Parameters

Table 26 reports the baseline parameter values used to generate Figure 8. Parameters that are not varied are held at these baseline values.

Table 26: Baseline parameter values used for comparative statics simulations

Parameter	Description	Baseline Value
σ	Std. dev. of the innovation shock ε_i	1.0
$\bar{a}$	Clinical quality threshold (P&T approval bar)	0.5
$\bar{p}$	Regulatory price cap	10.0
$\bar{c}$	Expected marginal cost	2.0
k	Effort cost scaling parameter	1.0
α_c	Rebate scale parameter	0.02
β_c	Rebate elasticity under power form $\tau_c = \alpha_c M_c^{\beta_c}$	0.75
c	Marginal cost distribution in the duopoly auction	$U[0.5, 3.5]$
M_0	Reference market size	100
δ	Period 2 cost reduction factor	0.7